

Support reciprocity and the q -zeta polynomial of preorder lattice-point posets

Two independent proofs, every negative q -integer, and arbitrary height functions

Merged research draft prepared for Vladimir Reshetnikov
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Abstract

Athanasiadis and Chapoton conjecture a negative-evaluation formula for the q -zeta polynomial of the coordinatewise poset of lattice points in a preorder polytope. This draft gives a proposed proof of the full q -identity, not only its unweighted specialization, and strengthens it to a separate identity for every support of a maximal lattice point. The strengthened statement is proved *twice*, by two arguments that share nothing beyond two classical imports and the elementary structure of the poset. The first proof runs through weighted incidence inversion, a support-restricted bipartite graph, and Ehrhart–Macdonald reciprocity at the capacities $\mathbf{1} + \mathbf{1}_T$; inside it, the geometric step is available in two interchangeable forms, one importing Postnikov’s bipartite duality and one deriving the duality it needs from a single application-free Euler identity for arbitrary bipartite graphs. The second proof uses no incidence matrix, no matching theorem and no second polytope: it keeps an auxiliary capacity alive through multivariate Ehrhart reciprocity, kills every non-maximal layer by a single specialization, and converts the result into a coefficient-transfer theorem for the involution $\mathcal{T}f(z) = (1 - z)f(-z/(1 - z))$. Only the second proof reaches every negative q -integer $[-r]_q$, where it produces manifestly nonnegative integer coefficients with a colored-composition meaning; only the first reaches arbitrary normalized height functions. We also obtain multivariate, transversal-matroid and Tutte interpretations, a characterization of the supports that occur, sharp extreme degrees, explicit coefficient formulas, worked examples, and three obstructions that bound the theorem in three different directions. Three independent exact verifiers are included. The proofs are mathematical, with the classical inputs stated explicitly; the computations are corroboration rather than a proof substitute.

Status. This is an AI-assisted research draft containing complete proposed arguments. It has not been independently refereed or verified in a proof assistant. The source checked still labels the target as Conjecture 4.9; focused literature searches found no separate resolution. This is not a certification of priority. Where the three source drafts merged here recorded a caveat differently, the most cautious wording has been kept.

Contents

1	The problem, the results, and the nonduplication boundary	4
1.1	What is proved	4
1.2	Two proofs, and two engines inside the first	5
1.3	What is deliberately kept more than once	6
1.4	Provenance	7

1.5	Relation to the other packages in this collection	7
2	Preorder polytopes as transport polytopes	8
3	Negative evaluation by weighted incidence inversion	10
3.1	An algebraic continuation, not an analytic one	10
3.2	Boolean localization	11
4	The classical polyhedral inputs, with conventions fixed	12
4.1	Hall’s theorem	12
4.2	The lattice-count formula	12
4.3	Bipartite draconian sequences and their duality	13
4.4	Ehrhart–Macdonald reciprocity	13
5	Independent capacities and their reciprocity	14
6	Proof A, Engine 1: the support–interior identity	15
7	Proof A, completion: from negative capacities to exact supports	17
7.1	The proof compressed into a chain of equalities	18
7.2	Ordinary reciprocity is only the total-count shadow	18
8	Proof A, Engine 2: a private-element Euler identity	18
8.1	An inequality description for sums of truncated simplices	19
8.2	Private elements and their alternating count	19
8.3	Residual matchings and privatization	21
9	Proof B: an involution transferring local weights	22
9.1	The $v = 0$ pivot	22
9.2	The involution and the transfer theorem	23
9.3	Proof B of the support identity	23
9.4	The q -zeta polynomial at every integer	24
9.5	A shorter derivation once capacity reciprocity is known	25
10	All negative evaluations: positivity and combinatorics	25
10.1	A colored-composition interpretation	26
10.2	The ordinary specialization at every r	26
10.3	Sharp extreme degrees	26
11	Consequences: matroids, feasible supports, and coefficients	27
11.1	Two matroid invariants for the same coefficient	27
11.2	Exactly which supports occur	28
11.3	Coefficients of the support polynomial	28
11.4	Products and an inclusion–exclusion algorithm	29
11.5	Arbitrary heights and a multivariate matrix identity	29

12 Explicit examples	30
12.1 Antichains, discrete preorders, and a single equivalence class	30
12.2 Chains	30
12.3 A fork, and why the direction of the preorder matters	31
12.4 A diamond	31
12.5 The five-element example in the source	32
13 Why the hypotheses cannot simply be discarded	32
13.1 Purity alone is insufficient: a pure downset, two failures	32
13.2 A reflexive, nontransitive incidence system	33
13.3 Arbitrary positive capacities do not satisfy the same formula	34
13.4 What the result does not settle	34
14 Exact computation and reproducibility	34
14.1 Exhaustive enumeration	35
14.2 Additional cases and independent routes	36
14.3 Two finite algorithms, and what they cost	37
14.4 Running and reusing the programs	37
15 Proof audit and research status	38
15.1 Dependencies	38
15.2 Normalization errors the proofs are sensitive to	38
15.3 What is established, and what is not	39
15.4 Provenance of the source discrepancy	40
16 Conclusion	40
A The finite Hall theorem used in Proof A	40
B A compact verification specification	41
B.1 Support, Boolean sums, and interior counts	41
B.2 The independent q -check	41
B.3 Capacities, transfer, and negative evaluations	42
C Compact proof outlines for independent checking	42
D Source and version ledger	42
E Suggested manifest entry	44

1 The problem, the results, and the nonduplication boundary

Let E be a finite set of cardinality n , and let τ be a preorder on E : a reflexive, transitive relation, with antisymmetry *not* required. An *ideal* is a set $I \subseteq E$ such that $j \in I$ and $i \leq_\tau j$ imply $i \in I$; a *filter* is the complement of an ideal, equivalently an upward closed set. For a vector x , write $x(I) = \sum_{i \in I} x_i$ and $|x| = x(E)$. Define

$$Q_\tau = \{x \in \mathbb{R}_{\geq 0}^E : x(I) \leq |I| \text{ for every ideal } I\}, \quad P_\tau = Q_\tau \cap \mathbb{Z}^E. \quad (1.1)$$

The order on P_τ is coordinatewise. Its rank is $|a|$, and all maximal points have rank n ; proofs of these basic facts are included below.

Put $[m]_q = (q^m - 1)/(q - 1)$. The q -zeta polynomial is characterized by

$$\mathcal{Z}_q(P, [m]_q) = \sum_{a_1 \leq \dots \leq a_{m-1}} q^{|a_1| + \dots + |a_{m-1}|} \quad (m \geq 2). \quad (1.2)$$

Here the chains are weak chains, repetitions are allowed, and there are $m - 1$ entries. The second argument of $\mathcal{Z}_q$ is an ordinary polynomial variable over $\mathbb{Q}(q)$, not a count of chain entries. In particular, $[-1]_q = -q^{-1}$. These conventions are those in Athanasiadis–Chapoton [1, Section 4.4], following Chapoton [2].

The target is Conjecture 4.9, equation (21), in [1]:

$$\boxed{\mathcal{Z}_q(P_\tau, -q^{-1}) = (-1)^n \sum_{a \in \text{Max}(P_\tau)} q^{-\text{cov}(a)}. \quad (1.3)}$$

The statistic $\text{cov}(a)$ counts the elements covered by a , not those covering a . For our coordinate downset,

$$\text{cov}(a) = |\text{supp}(a)|, \quad \text{supp}(a) = \{i : a_i > 0\}. \quad (1.4)$$

It is this identity (1.4), proved in Section 2, that makes (1.3) literally the source conjecture rather than a renormalized variant of it.

1.1 What is proved

For independent indeterminates $(z_i)_{i \in E}$, set

$$H_\tau(\mathbf{z}) = \sum_{a \in \text{Max}(P_\tau)} \prod_{i \in \text{supp}(a)} z_i, \quad \mathbf{z}^B = \prod_{i \in B} z_i, \quad (1.5)$$

and write $H_\tau(u) = H_\tau(u, \dots, u)$ for its diagonal. The polynomial H_τ is multiaffine: each variable has degree at most one. Different maximal points may contribute the same monomial. For $B \subseteq E$ put

$$N_\tau(B) = \#\{a \in \text{Max}(P_\tau) : \text{supp}(a) = B\}, \quad (1.6)$$

$$\eta_\tau(B) = \sum_{\substack{S \subseteq E \\ \mathbf{1}_B + \mathbf{1}_S \in P_\tau}} (-1)^{|S|}. \quad (1.7)$$

Theorem 1.1 (Support-refined reciprocity). *For every finite preorder τ on E ,*

$$H_\tau(\mathbf{z}) = (-1)^n \sum_{\substack{B, S \subseteq E \\ \mathbf{1}_B + \mathbf{1}_S \in P_\tau}} (-1)^{|B| + |S|} \mathbf{z}^B. \quad (1.8)$$

Equivalently, for every $B \subseteq E$,

$$N_\tau(B) = (-1)^{n - |B|} \eta_\tau(B) = (-1)^{n - |B|} \sum_{\substack{S \subseteq E \\ \mathbf{1}_B + \mathbf{1}_S \in P_\tau}} (-1)^{|S|}. \quad (1.9)$$

Conjecture (1.3) follows by setting every $z_i = q^{-1}$ and using weighted incidence inversion.

The second headline statement goes past $[-1]_q$. For integers $r \geq 1$ and $a \geq 0$, define

$$C_{r,0}(t) = 1, \quad C_{r,a}(t) = \sum_{s=1}^{\min(r,a)} \binom{a-1}{s-1} e_s(t, t^2, \dots, t^r) \quad (a \geq 1), \quad (1.10)$$

where e_s is the elementary symmetric polynomial of degree s . Every coefficient of $C_{r,a}$ is a nonnegative integer.

Theorem 1.2 (All negative q -integers). *For every finite preorder τ and every integer $r \geq 1$,*

$$(-1)^n \mathcal{Z}_q(P_\tau, [-r]_q) = \sum_{b \in \text{Max}(P_\tau)} \prod_{i \in E} C_{r,b_i}(q^{-1}) \in \mathbb{Z}_{\geq 0}[q^{-1}]. \quad (1.11)$$

For $n > 0$, the polynomial on the right, in the variable $t = q^{-1}$, is monic of degree rn ; its smallest exponent is the number c_τ of maximal equivalence classes of τ , and the coefficient there is the number of maximal points of that support size.

Here an equivalence class is a class of the relation $i \sim j$ if $i \leq_\tau j \leq_\tau i$; the quotient is a finite poset. Equivalence classes retain their original cardinalities throughout: replacing a class by a single element and forgetting that cardinality would change the problem. At $r = 1$ we have $C_{1,a}(t) = t$ for $a \geq 1$, so Theorem 1.2 contains (1.3).

The third headline statement goes sideways instead, replacing the rank by an arbitrary height statistic. Let $h : P_\tau \rightarrow \mathbb{N}$ be strictly increasing on comparable distinct elements with $h(\mathbf{0}) = 0$, and let $\mathcal{Z}_{q,h}$ be the height-weighted q -zeta polynomial of [2], defined by (1.2) with $|a_j|$ replaced by $h(a_j)$.

Theorem 1.3 (Arbitrary normalized heights). *For every such h ,*

$$\mathcal{Z}_{q,h}(P_\tau, [-1]_q) = (-1)^n \sum_{b \in \text{Max}(P_\tau)} q^{-h(\mathbf{1}_{\text{supp}(b)})}. \quad (1.12)$$

The exponent uses the Boolean support vector $\mathbf{1}_{\text{supp}(b)}$, not $h(b)$.

1.2 Two proofs, and two engines inside the first

Theorem 1.1 is proved twice. The two proofs are recorded in full, and deliberately so: the result is unrefereed, the arguments are independent, and each reaches a generalization the other does not.

Proof A (incidence inversion plus geometry), Sections 3–8. Weighted incidence-matrix inversion over $\mathbb{Q}(q)$ reduces the negative evaluation to points with coordinates in $\{0, 1, 2\}$ and produces the signed Boolean sums $\eta_\tau(B)$; this step works for an arbitrary finite coordinate downset and for an arbitrary normalized height, which is where Theorem 1.3 comes from. A geometric argument then evaluates those signed sums. Two interchangeable engines are available for it:

- **Engine 1 (Section 6), cumulative supports.** A support-restricted bipartite graph G_T and Postnikov's bipartite draconian duality identify maximal points supported inside T with the interior lattice points of $Q_{\tau^*}(\mathbf{1} + \mathbf{1}_T)$; Ehrhart–Macdonald reciprocity and a coordinatewise two-value interpolation finish. Shortest route, largest imported surface.
- **Engine 2 (Section 8), exact supports.** Two Hall lemmas reduce $\eta_\tau(B)$ at the *exact* support B to an alternating count over the independent sets of a private-element bipartite graph, and a standalone Euler identity, proved for *every* bipartite graph from two applications of Postnikov's lattice-count theorem plus Ehrhart, evaluates that count. Longest route, smallest imported surface: it re-derives, in the case it needs, the duality Engine 1 imports.

Proof B (the transfer involution), Section 9. No incidence matrix, no Hall theorem, no bipartite graph, no second polytope, no interior-point count. Instead: build the independent-capacity enumerator $\mathcal{E}_\sigma(u, v)$ and prove $\mathcal{E}_\sigma(-u, -v) = (-1)^n \mathcal{E}_\sigma(u-1, v-1)$ by interior translation and Ehrhart; then set $v = 0$, so that the auxiliary capacity slides from 0 to -1 and the binomial factor $\binom{n-|a|-1}{n-|a|}$ annihilates every non-maximal layer; then convert the resulting binomial identity into a transfer theorem for arbitrary local weights, through the involution $\mathcal{T}f(z) = (1-z)f(-z/(1-z))$ on the basis $(1-z)^m$; finally evaluate $\mathcal{Z}_q$ by an explicit Gaussian endpoint formula valid at *every* integer m and feed $f(z) = (1-z)(1-q^{-1}z)$ through $\mathcal{T}$. This is the only route that reaches $[-r]_q$ for every r , so it is where Theorem 1.2 comes from.

The two proofs share the two classical imports (Postnikov’s lattice-count theorem and Ehrhart–Macdonald reciprocity), the elementary structure of P_τ in Section 2, and nothing else. They also touch at two seams, which Sections 9 and 11 point out explicitly: the two-capacity collapse of Proof B is the two-value interpolation table of Proof A in binomial rather than polynomial language, and the independent-set family of Engine 2 is the family $\{S : \mathbf{1}_B + \mathbf{1}_S \in P_\tau\}$ of Engine 1, which is a transversal matroid.

1.3 What is deliberately kept more than once

Several things below exist in two or three versions. None of the duplications is an accident of merging: each buys something the others do not, and the place where each lives is listed here once and for all.

- (1) **Two proofs of Theorem 1.1** (Sections 3–8 and Section 9): they cross-validate an unrefereed result, and only one of them generalizes to $[-r]_q$ while only the other reaches arbitrary height functions.
- (2) **Two geometric engines inside Proof A** (Sections 6 and 8): Engine 1 is much shorter, Engine 2 imports strictly less, so a reader who declines Imported theorem 4.2 still has a complete route.
- (3) **Three derivations of the exact-support identity (1.9)** (Sections 7, 8 and 9.3): this is the statement the whole document is built on, so every independent route to it is worth having.
- (4) **Two constructions of the q -zeta polynomial at negative arguments** (Lemma 3.1 and Proposition 9.4): the matrix continuation is what makes arbitrary heights reachable, the endpoint formula is what makes every $[-r]_q$ reachable, and neither implies the other (Remark 9.5).
- (5) **Three obstructions carrying four failure statements** (Section 13): they bound purity, transitivity and unit capacities separately, and collapsing any of them would lose a boundary.
- (6) **Two matroid invariants for the same coefficient** (Proposition 11.1 and (11.3)): the Tutte evaluation places it inside a standard formalism, the independence-complex h -polynomial needs no matroid theorem at all.
- (7) **Three verifiers** (Section 14): they test three different theorems, at two different seeds and two different sampling depths, so the sampled populations are genuinely independent.
- (8) **Two statements about which supports occur** (Proposition 11.2 and Lemma 10.1 with Proposition 10.2): the qualitative cofinality criterion settles positivity, the quantitative extremal statement is what the degree claim of Theorem 1.2 depends on.
- (9) **Two sets of coefficient formulas** (Proposition 11.3 and Section 11): the third-from-the-top coefficient (11.5) and the product rule over disjoint unions came from different arguments and neither follows from the other.

- (10) **Two accounts of the $q \rightarrow 1$ specialization** (Section 7.2): the nilpotent argument is the cleaner justification, the interpolation is what the verifier implements, and both are needed to explain why the case already covered by [9] falls out safely rather than being reproved.
- (11) **Three reciprocity references** (Section 4.4): historical, textbook and cone-reciprocity, for readers arriving from three different literatures.
- (12) **Two seams between the two proofs** (Remarks 7.2 and 8.7): the two-capacity collapse against the two-value table, and Engine 2's independent sets against Engine 1's transversal matroid; recording them is what keeps the coincidences from looking like repetitions.

1.4 Provenance

This document is a merge of three separate research drafts prepared on 20 September 2026, each of which independently proposed a proof of (1.3) *and*, letter for letter, of the exact-support strengthening (1.9). All three are directories of the same collection:

enumerative-combinatorics/preorder-q-zeta-support-reciprocity

The spine of the present document. Proved (1.8) in its multivariate form by weighted incidence inversion, the support–interior identity at capacities $\mathbf{1} + \mathbf{1}_T$ through Postnikov's bipartite duality, univariate Ehrhart reciprocity along each capacity ray, a two-value interpolation table and a Bernstein-type collapse. Contributed Engine 1, the Tutte evaluation, the product rule, the nontransitive and capacity-two obstructions, and most of the package infrastructure.

enumerative-combinatorics/preorder-q-zeta-supportwise

Proved (1.9) at the exact support by residual and privatization Hall lemmas, followed by a self-contained Euler identity for arbitrary bipartite graphs obtained from two applications of Postnikov's Theorem 11.3. Contributed Engine 2, the arbitrary-height theorem, the $[t^{n-2}]$ coefficient formula, the finite Hall proof, the standalone bipartite test suite, and the pure-downset obstruction in its supportwise form.

enumerative-combinatorics/preorder-q-reciprocity-weighted

Proved (1.3) and (1.9) by independent-capacity reciprocity, the $v = 0$ pivot, and the coefficient-transfer involution, with no geometric duality and no matching theorem at all. Contributed Proof B, Theorem 1.2 and the whole $[-r]_q$ apparatus, the Gaussian endpoint formula, the colored-composition model, the sharp extreme degrees, and the pure-downset obstruction in its transfer form.

The merge states the shared theorem once, keeps both proofs, keeps both engines inside Proof A, and keeps every statement that exists in only one of the three. Section 15 records which caveat came from where, and Appendix D merges the three source ledgers.

1.5 Relation to the other packages in this collection

The supplied manifest [10] includes a package entitled *Reciprocity and Matrix Duality for Preorder Polytopes*, in the directory **enumerative-combinatorics/preorder-polytope-reciprocity**; we cite it as [9]. Its description explicitly claims the ordinary $q = 1$ case of Conjecture 4.9 and explicitly excludes the full q -refinement. The present target is exactly that excluded refinement. The manifest also contains results about other conjectures of the same source, and a separate entry, *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, which concerns a different conjecture and is not repeated here.

The difference is substantive. At $q = 1$, the right side of (1.3) retains only the total number of maximal points. At general q , it records the entire distribution of their support sizes.

Theorem 1.1 retains the actual support set, which is finer still, and Theorem 1.2 leaves the point $[-1]_q$ altogether.

Three specific overlaps with [9] are acknowledged rather than concealed, and are flagged again at the places where they occur.

- (i) The independent-capacity reciprocity of Proposition 5.4 is the identity already recorded in the manifest entry for [9], whose Theorem on multivariate reciprocity states it. We reprove it here in order to make Proof B independent of an unrefereed draft, not because it is claimed as new.
- (ii) The enumerator formula (5.3) for L_{τ^*} is the specialization at $s = 0$ of the rising-factorial enumerator of [9], derived there the same way from Postnikov's theorem and the filter identification of draconian vectors, with both drafts crediting [1, Lemma 4.1] for that identification. It is re-derived here and is not claimed as a new theorem.
- (iii) The ordinary case $\mathcal{Z}(P_{\tau}, -1) = (-1)^n \# \text{Max}(P_{\tau})$, recovered in Section 7.2, is the corollary already proved in [9]. We continue to present it as covered there and not as a fresh result; what is new here is the q -refinement above it and the support refinement beyond that.

No proof in [9] is imported as a lemma, and no uninspected proof in that package is assumed correct. Conversely, nothing here supersedes it. Only the manifest, and not the full text of the earlier packages, was available to the three drafts when they assessed nonduplication; that limitation is retained as stated.

Finally, a nonduplication boundary inside the source paper itself: H_{τ} counts *maximal* points by support, whereas the preorder h -polynomial of [1, Section 5] counts support sizes over *all* lattice points. These are different polynomials, and no claim is made about the latter.

There are three explicitly identified classical ingredients: the integral bipartite matching criterion (Hall's theorem, proved in Appendix A), Ehrhart–Macdonald reciprocity, and Postnikov's lattice-count and bipartite-duality theorems. The steps relating these ingredients to support sets, and their combination with the negative q -evaluation, are proved in full. No conjecture about real-rootedness, unimodality, or the h -polynomials of preorder polytopes is used.

2 Preorder polytopes as transport polytopes

For $J \subseteq E$, let

$$D_{\tau}(J) = \{i : i \leq_{\tau} j \text{ for some } j \in J\}, \quad U_{\tau}(J) = \{i : j \leq_{\tau} i \text{ for some } j \in J\}.$$

Transitivity makes these an ideal and a filter, respectively. Reflexivity gives $J \subseteq D_{\tau}(J)$ and $J \subseteq U_{\tau}(J)$. We abbreviate the closure of a singleton in the usual way. The dual preorder τ^* reverses all comparisons, so that the ideals of τ^* are the filters of τ .

For a nonnegative capacity vector $\mathbf{t} = (t_i)$ and a nonnegative scalar v , define

$$\begin{aligned} Q_{\tau}(\mathbf{t}) &= \{x \geq 0 : x(I) \leq t(I) \text{ for every ideal } I\}, \\ K_{\tau}(\mathbf{t}, v) &= \{x \geq 0 : x(I) \leq t(I) + v \text{ for every nonempty ideal } I\}. \end{aligned} \tag{2.1}$$

Thus $Q_{\tau} = Q_{\tau}(\mathbf{1})$ and $Q_{\tau}(\mathbf{t}) = K_{\tau}(\mathbf{t}, 0)$. The auxiliary capacity v plays no role in Proof A; it is the whole mechanism of Proof B, and it is introduced here once so that both proofs can use the same Minkowski realization.

Lemma 2.1 (Minkowski realization and transport description). *Write $\Delta_A^0 = \text{conv}(\{\mathbf{0}\} \cup \{e_j : j \in A\})$, so that $\Delta_{\emptyset}^0 = \{\mathbf{0}\}$. For all nonnegative real capacities,*

$$K_{\tau}(\mathbf{t}, v) = \sum_{i \in E} t_i \Delta_{U_{\tau}(i)}^0 + v \Delta_E^0. \tag{2.2}$$

For integral $\mathbf{t}$ and v , this is a lattice polytope. If $\mathbf{t} > 0$ coordinatewise, then $Q_\tau(\mathbf{t}) = K_\tau(\mathbf{t}, 0)$ has dimension n ; so does $K_\tau(\mathbf{t}, v)$ whenever $v > 0$.

Moreover $Q_\tau(\mathbf{t})$ is the transport polytope in which each $i \in E$ is a source supplying at most t_i units, each $j \in E$ is a destination requiring x_j units, transport from i to j is allowed exactly when $i \leq_\tau j$, and unused supply is permitted; equivalently,

$$x \in Q_\tau(\mathbf{t}) \iff x \geq 0 \text{ and } x(J) \leq t(D_\tau(J)) \text{ for every } J \subseteq E. \quad (2.3)$$

Proof. Denote the Minkowski sum in (2.2) by M . A summand $t_i \Delta_{U_\tau(i)}^0$ contributes at most t_i to an ideal I , and contributes zero if $i \notin I$, since an element of $U_\tau(i) \cap I$ would force $i \in I$. The last summand contributes at most v . Hence $M \subseteq K_\tau(\mathbf{t}, v)$.

For the reverse inclusion, fix $x \in K_\tau(\mathbf{t}, v)$ and a linear functional $w \in \mathbb{R}^E$. Put $w_j^+ = \max(w_j, 0)$ and $J_s = \{j : w_j \geq s\}$ for $s > 0$. When J_s is nonempty, its downward closure $D_\tau(J_s)$ is a nonempty ideal, so $x(J_s) \leq x(D_\tau(J_s)) \leq t(D_\tau(J_s)) + v$. Integrating this step-function inequality,

$$w \cdot x \leq w^+ \cdot x = \int_0^\infty x(J_s) ds \leq \sum_{i \in E} t_i \max(0, \max_{j \in U_\tau(i)} w_j) + v \max(0, \max_{j \in E} w_j),$$

and the right-hand side is exactly the support function of M at w . Both sets are compact, because nonnegativity and the constraint for $I = E$ bound all coordinates; the separation theorem gives $x \in M$. A finite Minkowski sum of integer multiples of lattice simplices is a lattice polytope. Reflexivity puts the box $\prod_i [0, t_i]$ inside $Q_\tau(\mathbf{t})$, which has nonempty interior when every $t_i > 0$; and $v \Delta_E^0$ is full-dimensional for $v > 0$.

For the transport description, any transport satisfies the ideal inequalities, because a unit received by a destination in an ideal I must have originated in I . Conversely, the capacitated bipartite matching criterion says that the demands can be met if and only if (2.3) holds: these are the source-capacity bounds for the neighbours of each set of destinations, that is, the cut conditions of the corresponding flow network. The ideal inequalities imply them, since $x(J) \leq x(D_\tau(J)) \leq t(D_\tau(J))$, and conversely every ideal I satisfies $D_\tau(I) = I$. The matching criterion itself is Hall's theorem, proved in Appendix A; the passage from vector inequalities to an ordinary finite matching replaces a demand of multiplicity x_j by x_j distinct copies, as explained there. $\square$

We restate the unit-capacity case in the language of allocations, because Proof A's Engine 2 uses it constantly.

Lemma 2.2 (Allocation criterion). *Think of each $r \in E$ as one available resource and of a_i as a number of demands of type i , a resource r being able to serve a demand of type i exactly when $r \leq_\tau i$. For $a \in \mathbb{N}^E$ the following are equivalent: (i) $a \in P_\tau$; (ii) the $|a|$ demands can be assigned injectively to resources along allowed edges; (iii) $a(A) \leq |D_\tau(A)|$ for every $A \subseteq E$.*

Proof. This is (2.3) at $\mathbf{t} = \mathbf{1}$, together with the integrality of the matching: (i) $\Rightarrow$ (iii) since $a(A) \leq a(D_\tau(A)) \leq |D_\tau(A)|$; (iii) $\Rightarrow$ (ii) by Hall's theorem applied to a_i distinct copies of each demand i , whose possible resources form $D_\tau(A)$ where A is the set of their types; and (ii) $\Rightarrow$ (i) because every demand whose type lies in an ideal I must use a resource in I . $\square$

Lemma 2.3 (Maximal points and intervals). *The set P_τ is a finite coordinate downset containing $\mathbf{0}$ and $\mathbf{1}$, indeed containing every $\mathbf{1}_S$. Its maximal elements are exactly its points of total coordinate sum n . Every interval $[a, b]$ is the coordinate box $\prod_i [a_i, b_i] \cap \mathbb{Z}^E$. In particular P_τ is graded with rank $|a|$, is pure of rank n , and (1.4) holds.*

Proof. Downward closure and the inclusion of the Boolean cube follow directly from the inequalities, since $|S \cap I| \leq |I|$. The inequality for $I = E$ gives $|a| \leq n$, proving finiteness. If a is integral

with $|a| < n$, choose an integral allocation as in Lemma 2.2; at least one resource r is unused, and adding one demand of type r served by r itself, which reflexivity permits, gives $a + e_r \in P_\tau$. Thus a is not maximal, while a point of total n cannot be increased in any coordinate.

If $a \leq c \leq b$ and $b \in P_\tau$, downward closure gives $c \in P_\tau$, which proves the interval assertion. Hence the elements covered by a are exactly the $a - e_i$ with $a_i > 0$, which is (1.4), and decrementing to $\mathbf{0}$ proves the rank formula. $\square$

Remark 2.4 (An alternative proof of the maximal layer). The maximal-layer statement can also be proved without any matching theorem, which matters for Proof B: if $a \in P_\tau$ has $|a| < n$ and $a + e_i \notin P_\tau$ for every i , then for each i some ideal $I_i \ni i$ is *tight*, that is $a(I_i) = |I_i|$. Modularity of $x \mapsto x(I)$ and of cardinality gives $a(I \cup J) + a(I \cap J) = |I \cup J| + |I \cap J|$ for tight I, J , and since each term on the left is at most the corresponding cardinality, both $I \cup J$ and $I \cap J$ are tight. The union of the I_i is E , so E is tight, contradicting $|a| < n$.

Remark 2.5 (Preorders are genuinely allowed). An equivalence class of τ is not contracted to a single coordinate. Its individual elements remain distinct unit-capacity sources and distinct destinations. Ideals contain a whole equivalence class or none of it, but support sets need not. This distinction is important both in the proofs and in the exhaustive tests. For the empty ground set, $P_\tau = \text{Max}(P_\tau) = \{()\}$ and all the polynomials below equal 1; the geometric statements are proved for $n \geq 1$ and this convention is used at $n = 0$.

3 Negative evaluation by weighted incidence inversion

This section is the first half of Proof A. Everything in it is valid for an arbitrary finite poset with least element and an arbitrary normalized height, hence in particular for any finite coordinate downset $P \subseteq \mathbb{N}^E$, not just a preorder polytope. It is a specialization of the q -incidence viewpoint of Chapoton [2, Appendix C], with the matrix conventions and the negative exponent verified explicitly.

3.1 An algebraic continuation, not an analytic one

Let P be a finite poset with least element $\mathbf{0}$, and let $h : P \rightarrow \mathbb{N}$ be *normalized*: strictly increasing on comparable distinct elements, with $h(\mathbf{0}) = 0$. Order the elements by nondecreasing h . Let Z be the incidence zeta matrix, let D be the weight matrix, and put

$$Z_{a,b} = \begin{cases} 1 & a \leq b, \\ 0 & \text{otherwise,} \end{cases} \quad D_{a,a} = q^{h(a)}, \quad A = ZD. \quad (3.1)$$

Use row vectors on the left: write $e_0^\top$ for the row supported at $\mathbf{0}$ and $\mathbf{j}$ for the column vector of ones; $\mathbf{j}$ is not an element of P . For every integer $\ell \geq 0$,

$$e_0^\top A^\ell \mathbf{j} = \sum_{a_1 \leq \dots \leq a_\ell} q^{h(a_1) + \dots + h(a_\ell)}, \quad (3.2)$$

the empty-chain sum at $\ell = 0$ being 1. Comparing with (1.2) at $\ell = m - 1$ identifies (3.2) with $Z_{q,h}(P, [m]_q)$ for $m \geq 2$.

Lemma 3.1 (Polynomial continuation to negative integers). *Let $g_0 < \dots < g_d$ be the distinct values of h . The expression $e_0^\top A^{m-1} \mathbf{j}$, for all integers m , is a polynomial of degree at most g_d in $[m]_q$, with coefficients in $\mathbb{Q}(q)$. Consequently*

$$Z_{q,h}(P, -q^{-1}) = e_0^\top A^{-2} \mathbf{j} = e_0^\top \mu D^{-1} \mu \mathbf{j}, \quad \mu = Z^{-1}. \quad (3.3)$$

Proof. Let V_j be the space of row vectors supported on elements of height at least g_j . Right multiplication by A preserves V_j , and on V_j/V_{j+1} it acts as the scalar q^{g_j} , because two comparable elements of the same height are equal. Therefore $A - q^{g_j}I$ maps V_j into V_{j+1} , and applying the factors in increasing order gives

$$\prod_{j=0}^d (A - q^{g_j}I) = 0.$$

Over $\mathbb{Q}(q)$ the factors are distinct, so A is diagonalizable with nonzero eigenvalues among $q^{g_0}, \dots, q^{g_d}$. Hence there are $c_j \in \mathbb{Q}(q)$ with

$$e_0^\top A^\ell \mathbf{j} = \sum_{j=0}^d c_j q^{g_j \ell} \quad \text{for every integer } \ell.$$

Put $H(X) = \sum_j c_j q^{-g_j} X^{g_j}$. For $m \geq 2$ and $\ell = m - 1$ this equals $H(q^m) = H(1 + (q - 1)[m]_q)$, and such a polynomial is unique because the $[m]_q$, $m \geq 2$, are infinitely many distinct elements of $\mathbb{Q}(q)$. So $\mathcal{Z}_{q,h}(P, t) = H(1 + (q - 1)t)$, and at $t = [-1]_q = -q^{-1}$ the argument of H is q^{-1} , that is $\ell = -2$: at $m = -1$ the matrix exponent is -2 , not -1 , precisely because the multichains in (1.2) have $m - 1$ entries. Finally $A^{-1} = D^{-1}\mu$ and $e_0^\top D^{-1} = e_0^\top$ because $h(\mathbf{0}) = 0$. No limiting or analytic-continuation assertion is involved anywhere in this argument; every step is an identity of polynomials over $\mathbb{Q}(q)$. $\square$

Remark 3.2. The argument just given also supplies, in this least-element setting, the polynomial existence that Chapoton's construction [2] requires; we do not assume it.

3.2 Boolean localization

From now on $P \subseteq \mathbb{N}^E$ is a finite coordinate downset and h is arbitrary normalized as above. Its intervals are products of integer chains by Lemma 2.3. A chain interval of positive length has Möbius number -1 at length one and 0 at greater length, directly from the defining recurrence $\sum_{a \leq c \leq b} \mu(a, c) = \delta_{a,b}$. Multiplying over coordinates,

$$\mu(a, b) = \begin{cases} (-1)^{|T|}, & b - a = \mathbf{1}_T \text{ for some } T \subseteq E, \\ 0, & \text{otherwise.} \end{cases} \quad (3.4)$$

In particular $\mu(\mathbf{0}, a)$ vanishes unless a is Boolean. This localization is why a statistic of full maximal vectors can be recovered from vectors with coordinates only 0, 1, 2.

Proposition 3.3 (The Boolean localization formula). *For every nonempty finite coordinate downset $P \subseteq \mathbb{N}^E$ and every normalized height h ,*

$$\mathcal{Z}_{q,h}(P, -q^{-1}) = \sum_{B \subseteq E} (-1)^{|B|} q^{-h(\mathbf{1}_B)} \eta_P(B), \quad \eta_P(B) = \sum_{\substack{S \subseteq E \\ \mathbf{1}_B + \mathbf{1}_S \in P}} (-1)^{|S|}. \quad (3.5)$$

For the ordinary rank $h = |\cdot|$ this reads

$$\mathcal{Z}_q(P, -q^{-1}) = \sum_{\substack{B, S \subseteq E \\ \mathbf{1}_B + \mathbf{1}_S \in P}} (-1)^{|B| + |S|} q^{-|B|}, \quad (3.6)$$

and, with $u = q^{-1}$, equivalently

$$\mathcal{Z}_q(P, -q^{-1}) = \sum_{c \in P \cap \{0,1,2\}^E} (-1)^{s(c)} (1 + u)^{s(c)} u^{d(c)}, \quad (3.7)$$

where $s(c) = \#\{i : c_i = 1\}$ and $d(c) = \#\{i : c_i = 2\}$.

Proof. Expand (3.3):

$$e_0^\top \mu D^{-1} \mu \mathbf{j} = \sum_{b \leq c} \mu(\mathbf{0}, b) q^{-h(b)} \mu(b, c) = \sum_{a \in P} \mu(\mathbf{0}, a) q^{-h(a)} \sum_{c \geq a} \mu(a, c).$$

By (3.4) a surviving b is $\mathbf{1}_B$ and a surviving $c - b$ is $\mathbf{1}_S$; conversely, if $\mathbf{1}_B + \mathbf{1}_S \in P$ then downward closure puts both points in the sum. This is (3.5), and (3.6) is its specialization. For (3.7), group the pairs (B, S) by the vector $c = \mathbf{1}_B + \mathbf{1}_S$: a coordinate of c equal to zero contributes 1; a coordinate equal to one arises in two ways and contributes $-1 - u$; a coordinate equal to two contributes u . Multiply over coordinates. $\square$

Thus (1.8) implies (1.3), since the two signs multiply to $(-1)^n$ and (1.4) identifies support size with the number of lower covers; and (1.8) together with (3.5) implies Theorem 1.3. What remains for Proof A is the evaluation of the signed Boolean sums $\eta_\tau(B) = \eta_{P_\tau}(B)$. Neither the matrix calculation nor the Boolean Möbius formula alone proves the desired sign or positivity; that is the work of Sections 4–8.

The rational function in (3.6) is a Laurent polynomial. Although the full poset may contain large coordinate values, this particular evaluation sees only its ternary-coordinate portion. This fact alone does *not* identify the answer with maximal points: Section 13 exhibits downsets where it is false. The special preorder geometry is still needed.

4 The classical polyhedral inputs, with conventions fixed

We state the precise versions used, so that no normalization of a trimmed polytope or of a draconian sequence remains implicit. These results are classical inputs, not contributions of this draft. Both proofs use Section 4.2 and Section 4.4; only Engine 1 of Proof A uses Section 4.3.

Throughout, for an indeterminate X and an integer $a \geq 0$,

$$\binom{X}{a} = \frac{X(X-1) \cdots (X-a+1)}{a!}, \quad \binom{X}{0} = 1, \quad (4.1)$$

$$f_a(X) = \binom{X+a-1}{a} = \frac{X(X+1) \cdots (X+a-1)}{a!}, \quad f_a(X+1) = \binom{X+a}{a}. \quad (4.2)$$

Every later evaluation with a negative upper argument means a polynomial evaluation, not the cardinality of a negatively sized set.

4.1 Hall's theorem

The finite marriage theorem [7] is used in Lemmas 2.1, 2.2, 8.2, 8.4, 8.5 and Proposition 11.1. It is stated and proved, with the exact matching convention and the multiplicity reduction fixed, in Appendix A. It is entirely absent from Proof B.

4.2 The lattice-count formula

Imported theorem 4.1 (Postnikov's untrimmed lattice-count formula). *Let F be a finite set with $|F| = d+1$, let $I_0 = F$, and let $I_1, \dots, I_m$ be nonempty subsets of F . Write $\Delta_I = \text{conv}\{e_j : j \in I\}$. For nonnegative integers $y_0, \dots, y_m$,*

$$\# \left(\left(\sum_{j=0}^m y_j \Delta_{I_j} \right) \cap \mathbb{Z}^F \right) = \sum_a f_{a_0}(y_0 + 1) \prod_{j=1}^m f_{a_j}(y_j), \quad (4.3)$$

where $a \in \mathbb{N}^{m+1}$ ranges over the vectors with

$$\sum_{j=0}^m a_j = d, \quad \sum_{j \in J} a_j \leq \left| \bigcup_{j \in J} I_j \right| - 1 \quad (\emptyset \neq J \subseteq \{0, \dots, m\}). \quad (4.4)$$

This is [3, Theorem 11.3] in its stated untrimmed form; compare [1, Theorem 2.1]. In Postnikov’s trimmed formulation, passing to the untrimmed polytope replaces the coefficient of the full simplex by one more than that coefficient, which is why the distinguished factor is $f_{a_0}(y_0 + 1) = \binom{y_0 + a_0}{a_0}$ and not $f_{a_0}(y_0)$. The theorem is used at zero weights as well, as its nonnegative-integer statement permits: an auxiliary full simplex may be introduced with $y_0 = 0$, and its factor is then $f_{a_0}(1) = 1$ for every $a_0 \geq 0$, not $\binom{a_0 - 1}{a_0}$. Forgetting this one-unit shift would invalidate every application below. The imported statement is Postnikov’s Theorem 11.3 together with its main proof; the passage of [3] explicitly called an “alternative semiproof” is not what is imported here.

4.3 Bipartite draconian sequences and their duality

Let $G = (L, R; \mathcal{E})$ be a connected bipartite graph with both parts nonempty. For $J \subseteq L$ write $N_G(J) \subseteq R$ for its neighbours, and define

$$\mathcal{D}_L(G) = \left\{ a \in \mathbb{N}^L : \sum_{i \in L} a_i = |R| - 1, \quad \sum_{i \in J} a_i \leq |N_G(J)| - 1 \quad (\emptyset \neq J \subseteq L) \right\}. \quad (4.5)$$

The set $\mathcal{D}_R(G)$ is defined by exchanging the parts.

Imported theorem 4.2 (Postnikov’s bipartite duality). *For such a graph, $|\mathcal{D}_L(G)| = |\mathcal{D}_R(G)|$.*

This is the draconian-sequence form of [3, Lemma 11.7 and Corollary 11.8]. One may view both sets as the lattice points of two dual trimmed generalized permutohedra; the triangulation interpretation explains why their cardinalities agree even when their ambient dimensions differ. Our application in Section 6 checks connectedness; it does not silently assume the theorem for a graph with isolated vertices.

Remark 4.3 (The import surface is a real trade). Imported theorem 4.2 is used only by Engine 1. Engine 2, in Section 8, proves the instance of this duality that it needs from Imported theorem 4.1 alone, applied twice to two polytopes in different ambient dimensions. Engine 1 is much shorter; Engine 2 rests on a strictly smaller imported surface. A reader who is unwilling to assume Lemma 11.7 of [3] still has a complete route to Theorem 1.1, and this is why both engines are written out.

4.4 Ehrhart–Macdonald reciprocity

For a full-dimensional lattice polytope $K \subseteq \mathbb{R}^d$, let $L_K(s) = \#(sK \cap \mathbb{Z}^d)$ for nonnegative integers s . Its polynomial continuation satisfies

$$L_K(-1) = (-1)^d \#(\text{int}(K) \cap \mathbb{Z}^d). \quad (4.6)$$

We use this standard reciprocity theorem in precisely this univariate form. The historical reference is Macdonald [4]; a textbook treatment is [5, Chapter 4]; and a primary-source proof of the more general rational-cone statement, from which the polytope case follows, is in Beck–Develin [6]. All three are retained because they are the historical, textbook and cone-reciprocity citations respectively; the three drafts merged here each used a different one. No assertion about a general multivariate reciprocity law, about changing normal fans, or about a weighted q -Ehrhart theorem is required anywhere below.

5 Independent capacities and their reciprocity

This section is shared by both proofs, and it is the only place where they share more than the two imports. Proof A needs it only at $v = 0$; Proof B needs the auxiliary capacity v to stay alive until after reciprocity has been applied, and then kills the non-maximal layers by setting it to zero. Setting $v = 0$ too early would hide exactly that operation.

Proposition 5.1 (The independent-capacity polynomial). *Let σ be a preorder on E . For all nonnegative integers u_i and v , the number of lattice points of $K_\sigma(\mathbf{u}, v)$ equals the evaluation at $(\mathbf{u}, v)$ of the polynomial*

$$\mathcal{E}_\sigma(\mathbf{u}, v) = \sum_{a \in P_{\sigma^*}} \binom{v+n-|a|}{n-|a|} \prod_{i \in E} f_{a_i}(u_i), \quad (5.1)$$

whose total degree is at most n .

Proof. Adjoin a new coordinate 0 and put $I_0 = E \cup \{0\}$ and $I_i = U_\sigma(i) \cup \{0\}$ for $i \in E$. Lift the right-hand side of (2.2) to

$$\widehat{K} = v \Delta_{I_0} + \sum_{i \in E} u_i \Delta_{I_i} \subseteq \mathbb{R}^{\{0\} \sqcup E}.$$

It lies in the affine hyperplane $x_0 + \sum_{i \in E} x_i = v + \sum_i u_i$, so projection forgetting x_0 is an affine lattice isomorphism from that hyperplane onto $\mathbb{R}^E$ and carries $\widehat{K}$ onto $K_\sigma(\mathbf{u}, v)$.

Apply Imported theorem 4.1 with $F = \{0\} \sqcup E$, so $d = n$. Any condition in (4.4) involving the index 0 is automatic, because I_0 is the full ground set. For nonempty $A \subseteq E$ the remaining condition is

$$a(A) \leq |U_\sigma(A)|, \quad U_\sigma(A) = \bigcup_{i \in A} U_\sigma(i). \quad (5.2)$$

If A is a filter this says $a(A) \leq |A|$; conversely those inequalities for all filters imply (5.2), since $a(A) \leq a(U_\sigma(A)) \leq |U_\sigma(A)|$. Filters of σ are ideals of σ^* , so the allowed a are exactly the points of P_{σ^*} , and the remaining coordinate $a_0 = n - |a|$ is determined. Substituting into (4.3) gives (5.1), the distinguished factor being $f_{a_0}(v+1) = \binom{v+a_0}{a_0}$. The degrees of the factors in each summand sum to n .

This filter identification of draconian vectors is the same one as [1, Lemma 4.1]; it is reproduced here to fix the dual orientation. $\square$

Proposition 5.2 (A weighted dual lattice-count polynomial). *Put $L_{\tau^*}(\mathbf{t}) = \mathcal{E}_{\tau^*}(\mathbf{t}, 0)$. Then*

$$\boxed{L_{\tau^*}(\mathbf{t}) = \sum_{a \in P_\tau} \prod_{i \in E} f_{a_i}(t_i)} \quad \text{and} \quad L_{\tau^*}(\mathbf{t}) = \#(Q_{\tau^*}(\mathbf{t}) \cap \mathbb{Z}^E) \quad (\mathbf{t} \in \mathbb{N}^E). \quad (5.3)$$

Proof. Take $\sigma = \tau^*$ in Proposition 5.1, so $\sigma^* = \tau$ and $K_{\tau^*}(\mathbf{t}, 0) = Q_{\tau^*}(\mathbf{t})$. The distinguished factor is $f_{a_0}(0+1) = 1$ for every $a_0 \geq 0$, including zero weights; the remaining factors are $f_{a_i}(t_i)$. The displayed formula also establishes polynomiality directly. $\square$

Remark 5.3 (Overlap with the earlier package, and why the dual appears). Formula (5.3) is the $s = 0$ specialization, with independent capacities, of the rising-factorial enumerator of [9], derived there in the same way from Postnikov's theorem and the same filter identification of draconian vectors, both drafts crediting [1, Lemma 4.1]. It is re-derived here so that this document is self-contained; it is not claimed as new. As for the orientation: in $Q_{\tau^*}(\mathbf{t})$ a source i may send to precisely the τ -lower elements $D_\tau(i)$, and these are the neighbourhoods that turn the draconian inequalities into the ideal inequalities defining P_τ . Using $U_\tau(i)$ at this point would interchange the preorder and its dual. This orientation is checked once more in Section 6.

Proposition 5.4 (Independent-capacity reciprocity). *The polynomials of (5.1) satisfy*

$$\mathcal{E}_\sigma(-\mathbf{u}, -v) = (-1)^n \mathcal{E}_\sigma(\mathbf{u} - \mathbf{1}, v - 1) \quad (5.4)$$

as an identity in all $n + 1$ independent variables.

Proof. First take all u_i and v to be positive integers. Then $K_\sigma(\mathbf{u}, v)$ is full-dimensional by Lemma 2.1, and $sK_\sigma(\mathbf{u}, v) = K_\sigma(s\mathbf{u}, sv)$ for $s \geq 0$ by homogeneity of the inequalities, so $L_K(s) = \mathcal{E}_\sigma(s\mathbf{u}, sv)$ as polynomials in s .

A lattice point x is interior precisely when

$$x_i \geq 1 \ (i \in E), \quad x(I) \leq u(I) + v - 1 \quad (\emptyset \neq I \text{ an ideal}).$$

Indeed, all the defining inequalities must be strict at an interior point, since equality in a valid inequality with nonzero normal prevents a full ball around it; and conversely simultaneous strictness of these finitely many inequalities gives an open neighbourhood inside the polytope, redundant inequalities causing no difficulty. The translation $x = z + \mathbf{1}$ is a bijection onto the integer points with

$$z \geq 0, \quad z(I) \leq (\mathbf{u} - \mathbf{1})(I) + (v - 1),$$

whose capacities are nonnegative. Now (4.6) and Proposition 5.1 give (5.4) for every positive integer capacity vector. Both sides are multivariate polynomials, and equality on a Cartesian product of infinite sets forces polynomial equality: fix all but one variable, use the univariate root bound, and repeat. $\square$

Remark 5.5 (Overlap with the earlier package). Independently weighted capacity reciprocity is exactly what the manifest entry for [9] already records, and its multivariate reciprocity theorem is (5.4). The proof is supplied here so that Proof B is independent of that unrefereed draft; the mechanism is credited to it as prior work within this collection and is not claimed as a contribution. What goes beyond it is the conversion of (5.4) into arbitrary local coefficient weights in Section 9, and the full q -evaluation.

Corollary 5.6 (Negative capacities as interior counts). *For every positive integral vector $\mathbf{c}$,*

$$L_{\tau^*}(-\mathbf{c}) = (-1)^n \#(\text{int } Q_{\tau^*}(\mathbf{c}) \cap \mathbb{Z}^E). \quad (5.5)$$

Proof. Fix $\mathbf{c} > 0$ integral. By homogeneity $Q_{\tau^*}(s\mathbf{c}) = sQ_{\tau^*}(\mathbf{c})$ for $s \geq 0$, and $Q_{\tau^*}(\mathbf{c})$ is a full-dimensional lattice polytope by Lemma 2.1. Hence $s \mapsto L_{\tau^*}(s\mathbf{c})$ is its Ehrhart polynomial. Evaluate this univariate polynomial at $s = -1$ and apply (4.6). $\square$

6 Proof A, Engine 1: the support–interior identity

This is the central combinatorial step of the first geometric engine. It works with *cumulative* supports. For $T \subseteq E$, put

$$B_\tau(T) = \#\{a \in \text{Max}(P_\tau) : \text{supp}(a) \subseteq T\}, \quad (6.1)$$

$$\Lambda_\tau(T) = \#(\text{int } Q_{\tau^*}(\mathbf{1} + \mathbf{1}_T) \cap \mathbb{Z}^E). \quad (6.2)$$

The capacity vector $\mathbf{1} + \mathbf{1}_T$ is the distinctive idea of this route: a capacity of 1 everywhere, raised to 2 exactly on T .

Theorem 6.1 (Restricted supports as interior lattice points). *For every $T \subseteq E$,*

$$\boxed{B_\tau(T) = \Lambda_\tau(T)}. \quad (6.3)$$

In particular,

$$B_\tau(T) = (-1)^n L_{\tau^*}(-\mathbf{1} - \mathbf{1}_T). \quad (6.4)$$

Proof. Assume first $n > 0$. If $T = \emptyset$, then $B_\tau(T) = 0$, since a maximal point has total n ; and an interior integral point of $Q_{\tau^*}(\mathbf{1})$ would have all coordinates at least one and total strictly less than n , which is impossible, so $\Lambda_\tau(T) = 0$ as well.

Let $T \neq \emptyset$. Form a bipartite graph G_T with left part T and right part $E \sqcup \{0\}$. Put an edge from t to 0 for every $t \in T$, and an edge from t to $i \in E$ exactly when

$$i \leq_\tau t. \quad (6.5)$$

For nonempty $J \subseteq T$ we then have $N_{G_T}(J) = \{0\} \cup D_\tau(J)$.

Left-hand draconian sequences. A vector $a \in \mathcal{D}_T(G_T)$ has total n and satisfies

$$a(J) \leq |D_\tau(J)| \quad (\emptyset \neq J \subseteq T). \quad (6.6)$$

Extend it by zero outside T . If $a \in P_\tau$, then (6.6) follows from the ideal inequalities. Conversely, for any ideal I let $J = T \cap I$. If $J = \emptyset$ then $a(I) = 0$; otherwise $D_\tau(J) \subseteq I$ and $a(I) = a(J) \leq |D_\tau(J)| \leq |I|$. Thus (6.6) is equivalent to membership in P_τ for vectors of total n supported in T , and Lemma 2.3 gives

$$|\mathcal{D}_T(G_T)| = B_\tau(T). \quad (6.7)$$

Right-hand draconian sequences. Write such a sequence as (d_0, y) with $y \in \mathbb{N}^E$; its total is $|T| - 1$. For nonempty $S \subseteq E$ its condition is

$$y(S) \leq |T \cap U_\tau(S)| - 1. \quad (6.8)$$

Every subset containing the right vertex 0 has all of T as its neighbour set, so those inequalities are automatic from the total and nonnegativity. Conditions (6.8) are equivalent to

$$y(I) \leq |T \cap I| - 1 \quad \text{for every nonempty } \tau\text{-filter } I : \quad (6.9)$$

for one direction take $S = I$, noting $U_\tau(I) = I$; for the other apply (6.9) to $U_\tau(S)$ and use $y(S) \leq y(U_\tau(S))$. Taking $I = E$ gives $|y| \leq |T| - 1$, so $d_0 = |T| - 1 - |y|$ is determined and nonnegative.

Since the ideals of τ^* are the filters of τ , the integral interior points of $Q_{\tau^*}(\mathbf{1} + \mathbf{1}_T)$ are exactly the integer vectors x with

$$x_i > 0 \ (i \in E), \quad x(I) < |I| + |T \cap I| \quad (\emptyset \neq I \text{ a } \tau\text{-filter}).$$

Full dimensionality was proved in Lemma 2.1; consequently strictness of these finitely many defining inequalities describes the ordinary interior, even when some of them are redundant. Substituting $x = y + \mathbf{1}$ turns these integral strict inequalities precisely into (6.9). We have therefore obtained a bijection $(d_0, y) \leftrightarrow y + \mathbf{1}$ between right-hand draconian sequences and interior points, giving

$$|\mathcal{D}_{E \sqcup \{0\}}(G_T)| = \Lambda_\tau(T). \quad (6.10)$$

Isolated vertices and the duality step. If $D_\tau(T) \neq E$, there is an isolated right vertex. In that case the left condition with $J = T$ would force $n \leq |D_\tau(T)| < n$, and the right condition for the singleton isolated vertex would force a nonnegative entry to be at most -1 ; both draconian sets are empty. If $D_\tau(T) = E$, every right vertex is adjacent to some left vertex, and the common neighbour 0 connects all left vertices, so G_T is connected and Imported theorem 4.2 applies. In both cases (6.7) and (6.10) yield (6.3).

For $n = 0$ use the usual zero-dimensional conventions: the only lattice point is $\mathbf{0}$, it is maximal and interior in $\mathbb{R}^0$, and both counts are one. Finally, (6.4) follows from Corollary 5.6 with $\mathbf{c} = \mathbf{1} + \mathbf{1}_T$. $\square$

The added vertex 0 serves two distinct and indispensable purposes. On the left it raises the total to n and cancels the “minus one” in the neighbourhood inequalities. On the right its entry stores the slack $|T| - 1 - |y|$, and the ordinary interior then produces exactly the strictness left over in the right neighbourhood inequalities.

7 Proof A, completion: from negative capacities to exact supports

For a finite coordinate downset $P \subseteq \mathbb{N}^E$, define the multiaffine polynomial

$$R_P(\mathbf{z}) = \sum_{\substack{B, S \subseteq E \\ \mathbf{1}_B + \mathbf{1}_S \in P}} (-1)^{|B|+|S|} \mathbf{z}^B = \sum_{B \subseteq E} (-1)^{|B|} \eta_P(B) \mathbf{z}^B. \quad (7.1)$$

The task is to prove $R_{P_\tau} = (-1)^n H_\tau$; by Proposition 3.3 this is Theorem 1.1.

Lemma 7.1 (Two-value interpolation). *For the weighted polynomial of (5.3),*

$$R_{P_\tau}(\mathbf{z}) = \sum_{T \subseteq E} \mathbf{z}^T \prod_{i \notin T} (1 - z_i) L_{\tau^*}(-\mathbf{1} - \mathbf{1}_T). \quad (7.2)$$

Proof. Insert (5.3) into the right-hand side and interchange the finite sums. The contribution of a fixed $a \in P_\tau$ factorizes as

$$\prod_{i \in E} ((1 - z_i) f_{a_i}(-1) + z_i f_{a_i}(-2)). \quad (7.3)$$

Directly from the polynomial definition (4.2) of f_k :

k	$f_k(-1)$	$f_k(-2)$	$(1 - z)f_k(-1) + zf_k(-2)$
0	1	1	1
1	-1	-2	$-(1 + z)$
2	0	1	z
$k \geq 3$	0	0	0

Thus only $a \in \{0, 1, 2\}^E$ survive. The three nonzero local factors are exactly the factors obtained by summing over all decompositions $a = \mathbf{1}_B + \mathbf{1}_S$ with sign $(-1)^{|B|+|S|}$ and weight $\mathbf{z}^B$. This proves (7.2). $\square$

Remark 7.2 (Seam: the same table in binomial language). The four-row table above is the same content as the two-capacity collapse of Section 9.5, which arises inside Proof B by combining the binomial identity (9.1) multilinearly over $u_i \in \{1, 2\}$: the local maximal-layer factor there becomes 1 at $b = 0$ and t at $b > 0$, exactly as the fourth column here becomes $1, -(1 + z), z, 0$ before the geometric step. The two derivations were found independently and in different languages, and the coincidence is one of the two seams where the two proofs touch. Neither is used in the other’s proof.

Proof A of Theorem 1.1. By Lemma 7.1 and Theorem 6.1,

$$\begin{aligned} R_{P_\tau}(\mathbf{z}) &= (-1)^n \sum_{T \subseteq E} \mathbf{z}^T \prod_{i \notin T} (1 - z_i) B_\tau(T) \\ &= (-1)^n \sum_{a \in \text{Max}(P_\tau)} \sum_{T \supseteq \text{supp}(a)} \mathbf{z}^T \prod_{i \notin T} (1 - z_i). \end{aligned}$$

For any fixed $B \subseteq E$,

$$\sum_{T \supseteq B} \mathbf{z}^T \prod_{i \notin T} (1 - z_i) = \mathbf{z}^B \prod_{i \notin B} (z_i + (1 - z_i)) = \mathbf{z}^B. \quad (7.4)$$

Consequently $R_{P_\tau} = (-1)^n H_\tau$, which is (1.8). Taking the coefficient of $\mathbf{z}^B$ proves (1.9); the exponents $n + |B|$ and $n - |B|$ have the same parity.

Finally, Proposition 3.3 identifies $\mathcal{Z}_q(P_\tau, -q^{-1})$ with $R_{P_\tau}(q^{-1}, \dots, q^{-1})$, and Lemma 2.3 identifies the number of lower covers with support size. This proves the full formula (1.3) as an identity in $\mathbb{Q}(q)$. Substituting (1.9) into (3.5) for a general normalized height proves Theorem 1.3. $\square$

7.1 The proof compressed into a chain of equalities

Writing $w_T(\mathbf{z}) = \mathbf{z}^T \prod_{i \notin T} (1 - z_i)$, Proof A is

$$\begin{aligned} R_{P_\tau}(\mathbf{z}) &= \sum_T w_T(\mathbf{z}) L_{\tau^*}(-\mathbf{1} - \mathbf{1}_T) && \text{by the local binomial table,} \\ &= (-1)^n \sum_T w_T(\mathbf{z}) \Lambda_\tau(T) && \text{by Ehrhart reciprocity,} \\ &= (-1)^n \sum_T w_T(\mathbf{z}) B_\tau(T) && \text{by bipartite duality,} \\ &= (-1)^n H_\tau(\mathbf{z}) && \text{by (7.4).} \end{aligned}$$

The q -zeta statement is the diagonal specialization, after the independent incidence calculation of Section 3. The argument is finite throughout: no convergence, no analytic continuation in a numerical parameter, and no exchange of infinite sums is involved.

7.2 Ordinary reciprocity is only the total-count shadow

At $q = 1$, positive evaluations specialize to ordinary multichain counts. Polynomial interpolation at the distinct nodes $m = 2, \dots, n+2$ shows that this specialization of the q -zeta polynomial is the ordinary zeta polynomial; this is the route the shipped verifier implements. A cleaner justification avoids any specialization of a rational function in q altogether: the ordinary zeta polynomial is represented by $e_0^\top Z^{m-1} \mathbf{j}$, and since $Z = I + N$ with N nilpotent, $Z^{m-1} = \sum_j \binom{m-1}{j} N^j$ is polynomial in m and evaluates to Z^{-2} at $m = -1$. Either way,

$$\mathcal{Z}(P_\tau, -1) = (-1)^n \# \text{Max}(P_\tau).$$

Both accounts are recorded because the merge must not appear to prove this case again, but must explain why it falls out safely: this is the case already claimed in the manifest and proved in [9], and it is not presented here as a new result.

Conversely, knowledge of this single total count cannot recover (1.9). The additional information is retained by the 2^n choices of the capacities $\mathbf{1} + \mathbf{1}_T$, and the two-value interpolation converts it into support coefficients.

8 Proof A, Engine 2: a private-element Euler identity

Engine 1 evaluated the cumulative counts $B_\tau(T)$ and recovered exact supports afterwards, at the cost of importing Postnikov's bipartite duality. Engine 2 attacks the exact support B directly. It costs two Hall lemmas – the point at which transitivity is spent – and a longer geometric argument, and it buys back the import: it uses Imported theorem 4.1 alone, applied twice, and thereby *proves*, in the case it needs, the duality that Engine 1 imports. Either engine completes Proof A on its own. Both are recorded, as announced in Remark 4.3.

8.1 An inequality description for sums of truncated simplices

Lemma 8.1. *Let $A_u \subseteq R$ for u in a finite index set U , where R is a finite coordinate set. Then*

$$\sum_{u \in U} \Delta_{A_u}^0 = \left\{ z \in \mathbb{R}_{\geq 0}^R : z(J) \leq \#\{u : A_u \cap J \neq \emptyset\} \text{ for all } J \subseteq R \right\}. \quad (8.1)$$

Proof. The inclusion from left to right holds separately for each summand. For the reverse inclusion let z satisfy the inequalities and let $w \in \mathbb{R}^R$. Replacing the negative entries of w by zero can only increase $w \cdot z$. Let the distinct positive entries be $0 < c_1 < \dots < c_k$, put $c_0 = 0$ and $J_j = \{i : w_i \geq c_j\}$. Then

$$w \cdot z \leq \sum_{j=1}^k (c_j - c_{j-1}) z(J_j) \leq \sum_{u \in U} \max(0, \max_{i \in A_u} w_i),$$

and the last expression is the support function of the Minkowski sum. It bounds $w \cdot z$ for every w , which places z in that compact convex set by the separating-hyperplane characterization. Empty A_u contribute zero to the support function. This is the general truncated-simplex-sum statement, for an arbitrary indexed family; Lemma 2.1 is the instance in which the family comes from a preorder. $\square$

8.2 Private elements and their alternating count

Let H be *any* bipartite graph with disjoint vertex classes R and C . Write $r = |R|$, $c = |C|$, and $N(i) \subseteq C$ for the neighbours of $i \in R$. Give each resource $i \in R$ an additional *private* ground element p_i , adjacent to resource i alone; the augmented ground set is

$$U = \{p_i : i \in R\} \sqcup C.$$

Let $\mathcal{I}_H$ consist of the subsets of U that can be matched injectively into R . Let $\text{Deg}(H)$ be the set of distinct vectors

$$d = (d_j)_{j \in C}, \quad d_j = \#\{i \in R : f(i) = j\}, \quad f(i) \in N(i), \quad (8.2)$$

that is, the set of degree vectors of full allocations – not the set of allocations themselves.

Lemma 8.2 (Private-element Euler identity). *For every bipartite graph H as above,*

$$(-1)^r \sum_{T \in \mathcal{I}_H} (-1)^{|T|} = |\text{Deg}(H)|. \quad (8.3)$$

Proof. If $R = \emptyset$, the only independent subset is empty and the unique full allocation has zero degree vector, so both sides are 1. If some $i \in R$ has $N(i) = \emptyset$, resource i can serve only p_i ; toggling p_i preserves membership in $\mathcal{I}_H$ and reverses the sign, so the left side is zero, while the right side is zero because a full allocation into C is impossible. We may therefore assume $r \geq 1$ and all $N(i)$ nonempty; in particular $c \geq 1$.

Step 1: turn the alternating sum into an Ehrhart value. For a ground element $u \in U$ let $A_u \subseteq R$ be its allowed resources, so $A_{p_i} = \{i\}$ and $A_j = \{i : j \in N(i)\}$ for $j \in C$. Set

$$K = \sum_{u \in U} \Delta_{A_u}^0 = [0, 1]^R + \sum_{j \in C} \Delta_{A_j}^0. \quad (8.4)$$

Introduce a coordinate 0 and homogenize each $\Delta_{A_u}^0$ to $\Delta_{\{0\} \cup A_u}$ on $F = \{0\} \sqcup R$. For a fixed nonnegative integer s , dropping the zeroth coordinate is an affine lattice bijection from the

corresponding sum of simplices onto sK , since the sum of all coordinates is fixed at $s|U|$. Introduce one further, full, simplex on F with weight zero.

In (4.4) a nonempty $V \subseteq U$ now imposes $a(V) \leq |\bigcup_{u \in V} A_u|$; subsets containing the full-simplex index give no additional condition, and its exponent is $a_0 = r - \sum_{u \in U} a_u$. Consequently Imported theorem 4.1 gives the polynomial identity

$$L_K(s) = \sum_{a \in \mathcal{A}} \prod_{u \in U} f_{a_u}(s), \quad (8.5)$$

$$\mathcal{A} = \left\{ a \in \mathbb{N}^U : a(V) \leq |\bigcup_{u \in V} A_u| \text{ for all } V \subseteq U \right\},$$

the sum being finite because $\sum_u a_u \leq r$. The equality holds for all nonnegative integers s and is therefore an equality of polynomials. At $s = -1$ a factor $f_{a_u}(-1)$ is 1 for $a_u = 0$, -1 for $a_u = 1$, and 0 for $a_u \geq 2$; Hall's theorem identifies the surviving Boolean vectors with $\mathcal{I}_H$. Thus

$$L_K(-1) = \sum_{T \in \mathcal{I}_H} (-1)^{|T|}. \quad (8.6)$$

Step 2: describe the interior lattice points. The cube in (8.4) makes K full-dimensional in $\mathbb{R}^R$. For $J \subseteq R$ write $N(J) = \bigcup_{i \in J} N(i)$. Lemma 8.1 gives

$$K = \{z \geq 0 : z(J) \leq |J| + |N(J)| \text{ for all } J \subseteq R\}.$$

Its interior is obtained by making the coordinate inequalities and all nonempty-subset inequalities strict: an interior point cannot lie on a proper supporting hyperplane, and conversely strict satisfaction of this finite system gives a neighbourhood inside K . For an integer point z put $y = z - \mathbf{1}_R$. Then

$$z \in \text{int } K \cap \mathbb{Z}^R \iff y \in \mathbb{N}^R \text{ and } y(J) \leq |N(J)| - 1 \quad (\emptyset \neq J \subseteq R). \quad (8.7)$$

The condition $J = R$ in particular bounds $\sum_i y_i$ by $c - 1$. Note that the shift removes the private cube and leaves *strict* neighbourhood bounds, that is $y(J) \leq |N(J)| - 1$ and not $y(J) \leq |N(J)|$.

Step 3: count the same vectors on the other side of the graph. Consider

$$P_H = \sum_{i \in R} \Delta_{N(i)} \subseteq \mathbb{R}^C.$$

Apply Imported theorem 4.1 once more, with a full simplex Δ_C of weight zero and each displayed summand of weight one. Every binomial factor then equals 1. The resulting exponent vectors are precisely the y of (8.7), completed by $a_0 = c - 1 - \sum_i y_i$; subsets containing the full-simplex index impose only the automatic bound $a(J) \leq c - 1$. Hence

$$\#(\text{int } K \cap \mathbb{Z}^R) = \#(P_H \cap \mathbb{Z}^C). \quad (8.8)$$

Every degree vector in (8.2) is a lattice point of P_H . Conversely, for $d \in P_H \cap \mathbb{Z}^C$ one has $\sum_j d_j = r$ and $d(A) \leq \#\{i : N(i) \cap A \neq \emptyset\}$ for all $A \subseteq C$; Hall's theorem, applied to d_j copies of each target j , provides an injective assignment of these r demands to the r resources. All resources are used, so this is a full allocation with degree vector d . Hence $P_H \cap \mathbb{Z}^C = \text{Deg}(H)$. Combining (8.6), Ehrhart reciprocity (4.6) and (8.8) proves (8.3). $\square$

Remark 8.3 (What the geometric argument does and does not assert). The two sets in (8.8) generally lie in different ambient dimensions. The proof is an equality of their cardinalities obtained from Imported theorem 4.1; it does not claim an affine bijection between them, and it does not count matchings with multiplicity. Both distinctions matter when the lemma is applied. Lemma 8.2 is stated and proved for *every* bipartite graph, not merely those arising from preorders, so it is a reusable graph-theoretic statement in its own right; the standalone test suite of Section 14 exercises it in that generality.

8.3 Residual matchings and privatization

Fix $B \subseteq E$ and put $R = E \setminus B$.

Lemma 8.4 (Subtracting a Boolean vector). *For $y \in \mathbb{N}^E$, the vector $\mathbf{1}_B + y$ lies in P_τ exactly when the demands specified by y can be assigned injectively to resources in R , using the relation $r \leq_\tau i$.*

Proof. The feasibility inequalities after subtraction are

$$y(I) \leq |I| - |I \cap B| = |I \cap R| \quad \text{for every ideal } I. \quad (8.9)$$

For any set A of demand types these imply $y(A) \leq y(D_\tau(A)) \leq |R \cap D_\tau(A)|$, and Hall's theorem gives the matching. Conversely such a matching uses only resources in R , while the base demands $\mathbf{1}_B$ can use their own resources in B ; the combined matching is feasible by Lemma 2.2. $\square$

The residual graph still has targets in R as well as targets in B : a resource r can serve a different target $u \in R$ when $r \leq_\tau u$. The next lemma removes those extra edges without changing any Boolean feasibility test. It is the point at which transitivity does its work in this engine, and it has no counterpart in Engine 1 or in Proof B.

Lemma 8.5 (Privatizing the residual targets). *For $U \subseteq R$ and $V \subseteq B$, the set of demands $U \sqcup V$ can be matched to R under $\leq_\tau$ if and only if V can be matched to $R \setminus U$ under $\leq_\tau$.*

Proof. The reverse implication follows by assigning each $u \in U$ to its own resource. For the forward implication let $A \subseteq V$ and set $I = D_\tau(A)$, an ideal by transitivity. Every matched demand in $(U \cup V) \cap I$ must use a resource in $R \cap I$, so

$$|A| + |U \cap I| \leq |(U \cup V) \cap I| \leq |R \cap I|, \quad \text{whence} \quad |A| \leq |(R \setminus U) \cap D_\tau(A)|.$$

This is Hall's condition for matching V to $R \setminus U$. No antisymmetry was used. $\square$

Now form the bipartite graph H_B with resource set $R = E \setminus B$, target set B , and edges

$$r H_B s \iff r \leq_\tau s, \quad (8.10)$$

and adjoin to it a private ground element for each resource, identified with the corresponding element of R . Lemmas 8.4 and 8.5 show that

$$\eta_\tau(B) = \sum_{T \in \mathcal{I}_{H_B}} (-1)^{|T|}. \quad (8.11)$$

Lemma 8.6 (Maxima with prescribed support). *The map $b \mapsto (b - \mathbf{1}_B)|_B$ is a bijection from the maximal elements of P_τ with support B onto $\text{Deg}(H_B)$.*

Proof. For such a b , Lemma 2.3 gives $|b| = n$. The nonnegative vector $y = b - \mathbf{1}_B$ is supported in B and has sum $n - |B| = |R|$; by Lemma 8.4 its demands can be assigned to R , and equality of the totals forces every resource to be used. Thus $y|_B$ is a full-allocation degree vector for H_B . Conversely a degree vector in $\text{Deg}(H_B)$, extended by zero outside B and added to $\mathbf{1}_B$, is feasible, has sum n and has support exactly B , hence is maximal. The two constructions are inverse on vectors; different allocations realizing the same degree vector produce the same b , which is exactly what is required, since the final objects are distinct degree vectors and not matchings counted with multiplicity. $\square$

Proof A of Theorem 1.1, via Engine 2. By (8.11) and Lemma 8.2,

$$(-1)^{n-|B|}\eta_\tau(B) = |\text{Deg}(H_B)| = N_\tau(B),$$

the last equality being Lemma 8.6. This is (1.9), and it includes $B = \emptyset$, $B = E$ and $E = \emptyset$ through the boundary cases of Lemma 8.2. Summing $(-1)^{|B|}\eta_\tau(B)\mathbf{z}^B$ over B gives (1.8), and substitution into (3.5) proves (1.3) and Theorem 1.3 as before. $\square$

Remark 8.7 (Seam: one family, two descriptions). The family $\mathcal{I}_{H_B}$ just used is precisely the family $\{S \subseteq E : \mathbf{1}_B + \mathbf{1}_S \in P_\tau\}$ that Engine 1 meets in (1.9), and Proposition 11.1 identifies it as the independent-set family of a transversal matroid. In Engine 2 that identification is load-bearing and is proved by Hall’s theorem before anything geometric happens; in Engine 1 it appears afterwards, as an interpretation. This is the second of the two seams announced in Section 1.2.

9 Proof B: an involution transferring local weights

Proof B is architecturally independent of Proof A end to end. It uses no incidence matrix, no Möbius function, no Hall theorem, no bipartite graph, no second polytope and no interior-point count. Its only inputs are Section 5 – itself resting on the same two classical theorems – and the elementary structure of P_τ . Keeping it is not redundancy: it cross-validates an unrefereed result, and it is the only route that reaches $[-r]_q$ for every r .

9.1 The $v = 0$ pivot

Proposition 9.1 (Maximal-layer binomial identity). *For independent indeterminates u_i ,*

$$\sum_{a \in P_\tau} \prod_{i \in E} \binom{a_i - u_i - 1}{a_i} = (-1)^n \sum_{b \in \text{Max}(P_\tau)} \prod_{i \in E} \binom{b_i + u_i - 2}{b_i}. \quad (9.1)$$

Proof. Apply Proposition 5.4 with $\sigma = \tau^*$, so that the index set of (5.1) is $P_{\sigma^*} = P_\tau$, and then set $v = 0$. On the left of (5.4) the first factor of (5.1) is $\binom{n-|a|}{n-|a|} = 1$. On the right it becomes

$$\binom{n-|a|-1}{n-|a|} = \begin{cases} 1, & |a| = n, \\ 0, & |a| < n, \end{cases}$$

which annihilates every layer below the top. This is a polynomial specialization, legitimate because reciprocity was first established in all $n+1$ variables. Lemma 2.3 identifies the surviving layer with $\text{Max}(P_\tau)$. The remaining factors are exactly those of (9.1), since $f_a(-u) = \binom{a-u-1}{a}$ and $f_b(u-1) = \binom{b+u-2}{b}$ directly from (4.2). $\square$

The source of the top-layer restriction is now transparent, and it is worth stating what it is not. The auxiliary capacity changes from 0 to -1 , annihilating every positive slack $n - |a|$. There is *no* geometric object “of capacity -1 ” anywhere in this argument, and no duality is invoked: the selection of the maximal layer is a pure algebraic specialization. This is the sharpest contrast with Proof A, where the same selection is performed by a bijection with interior lattice points (Engine 1) or with full allocations in a residual graph (Engine 2).

9.2 The involution and the transfer theorem

For a formal power series over a commutative $\mathbb{Q}$ -algebra, define

$$(\mathcal{T}f)(z) = (1 - z)f\left(\frac{-z}{1 - z}\right). \quad (9.2)$$

The substituted series has zero constant term, so the composition is well-defined. The substitution $h(z) = -z/(1 - z)$ satisfies $h(h(z)) = z$ and $1 - h(z) = 1/(1 - z)$; hence $\mathcal{T}^2 f = f$.

Theorem 9.2 (Coefficient transfer). *For any family $(f_i)_{i \in E}$ of formal power series over a commutative $\mathbb{Q}$ -algebra,*

$$\boxed{\sum_{a \in P_\tau} \prod_{i \in E} [z^{a_i}] f_i(z) = (-1)^n \sum_{b \in \text{Max}(P_\tau)} \prod_{i \in E} [z^{b_i}] (\mathcal{T}f_i)(z).} \quad (9.3)$$

Proof. All indices on either side are at most n , and the first $n + 1$ coefficients of $\mathcal{T}f$ depend only on the first $n + 1$ coefficients of f , so every f_i may be truncated after degree n . The polynomials $(1 - z)^m$, $0 \leq m \leq n$, form a basis of that polynomial space, and the identity is multilinear in the f_i , so it suffices to take $f_i = (1 - z)^{m_i}$ with $0 \leq m_i \leq n$. For this basis,

$$\begin{aligned} [z^a](1 - z)^m &= (-1)^a \binom{m}{a} = \binom{a - m - 1}{a}, & \mathcal{T}((1 - z)^m) &= (1 - z)^{1-m}, \\ [z^b](1 - z)^{1-m} &= \binom{b + m - 2}{b}, \end{aligned}$$

so that (9.3) is (9.1) with $u_i = m_i$. Multilinearity and truncation finish the proof. $\square$

Theorem 9.2 is stated for arbitrary families of formal power series over any commutative $\mathbb{Q}$ -algebra, which is far more general than the conjecture requires; that generality is what makes both Theorem 1.1 and Theorem 1.2 instances of one statement.

Remark 9.3 (A concrete coefficient rule). If $f(z) = \sum_{k \geq 0} f_k z^k$ and $\mathcal{T}f = \sum_{a \geq 0} g_a z^a$, then

$$g_0 = f_0, \quad g_1 = -f_0 - f_1, \quad g_a = \sum_{k=2}^a (-1)^k \binom{a-2}{k-2} f_k \quad (a \geq 2), \quad (9.4)$$

by expanding $\mathcal{T}(z^k) = (-1)^k z^k (1 - z)^{1-k}$. The transfer therefore uses a triangular integral matrix, and is particularly simple to test without a symbolic algebra system.

9.3 Proof B of the support identity

Proof B of Theorem 1.1. Take $f_i(z) = (1 - z)(1 - z_i z)$, whose coefficient sequence is $1, -(1 + z_i), z_i, 0, 0, \dots$, whereas

$$\mathcal{T}f_i(z) = 1 + \frac{z_i z}{1 - z}$$

has coefficient 1 in degree zero and z_i in every positive degree. The left side of (9.3) is then

$$\sum_{a \in P_\tau \cap \{0,1,2\}^E} \prod_{i: a_i=1} [-(1 + z_i)] \prod_{i: a_i=2} z_i = R_{P_\tau}(\mathbf{z}),$$

by grouping the pairs (B, S) of (7.1) according to $a = \mathbf{1}_B + \mathbf{1}_S$, and the right side is $(-1)^n \sum_b \prod_{i \in \text{supp}(b)} z_i = (-1)^n H_\tau(\mathbf{z})$. Hence $R_{P_\tau} = (-1)^n H_\tau$, which is (1.8), and taking the coefficient of $\mathbf{z}^B$ gives (1.9) in the form

$$N_\tau(B) = (-1)^{n+|B|} \sum_{S \subseteq E} (-1)^{|S|} \mathbf{1}_{B+\mathbf{1}_S \in P_\tau}. \quad (9.5)$$

$\square$

This is a third, independent derivation of the exact-support identity: via the transfer theorem, rather than via incidence inversion plus residual matchings (Engine 2) or via cumulative counts plus two-value interpolation (Engine 1). Since (1.9) is the strengthened statement on which this whole document is built, all three routes to it are worth recording.

Formula (9.5) counts maximal elements with specified support without enumerating their possibly large coordinate entries, and it displays a nontrivial positivity statement for an alternating sum of feasibility indicators. For $B = E$ the only feasible S is the empty set, so the coefficient of $\prod_i z_i$ is one, as expected from the unique all-positive vector of sum n .

9.4 The q -zeta polynomial at every integer

To complete Proof B we must connect (1.8) to $\mathcal{Z}_q$ without using the incidence calculation of Section 3. This is done by constructing the q -zeta polynomial outright. For $a \geq 0$ and an integer m use the rational-function convention

$$\left[\begin{matrix} m + a - 2 \\ a \end{matrix} \right]_q = \prod_{j=1}^a \frac{1 - q^{m+j-2}}{1 - q^j}, \quad (9.6)$$

with empty product one; it extends the usual Gaussian binomial to the negative upper arguments occurring below. The variable q is indeterminate throughout.

Proposition 9.4 (Gaussian endpoint formula). *For every integer m ,*

$$\mathcal{Z}_q(P_\tau, [m]_q) = \sum_{a \in P_\tau} q^{|a|} \prod_{i \in E} \left[\begin{matrix} m + a_i - 2 \\ a_i \end{matrix} \right]_q. \quad (9.7)$$

Proof. First let $m \geq 2$ and condition a multichain on its final element a . Every earlier point is coordinatewise at most a , hence lies in P_τ by the downset property, and the choices in the separate coordinates are independent. In a coordinate with endpoint a_i , the preceding $m - 2$ entries form a weakly increasing sequence in $\{0, \dots, a_i\}$, whose weight enumerator is $\left[\begin{matrix} m + a_i - 2 \\ a_i \end{matrix} \right]_q$; the endpoint contributes q^{a_i} . For an elementary verification of the local enumerator, let $R_{k,a}$ count weakly increasing k -tuples with entries between zero and a by the sum of their entries; separating according to whether an entry a occurs gives $R_{k,a} = R_{k,a-1} + q^a R_{k-1,a}$ with $R_{0,a} = R_{k,0} = 1$, and the Gaussian polynomial $\left[\begin{matrix} k + a \\ a \end{matrix} \right]_q$ satisfies the same recurrence and boundary values.

To justify the remaining integers, regard $Q = q^m$ as a new variable. Each local product in (9.6) is a polynomial in Q of degree at most a_i over $\mathbb{Q}(q)$, so the right side of (9.7) is a polynomial in Q of degree at most n . Replacing Q by $1 + (q - 1)X$ yields a polynomial in X whose values at $X = [m]_q$, $m \geq 2$, agree with (1.2). Such a polynomial is unique, because these arguments are infinitely many distinct elements of $\mathbb{Q}(q)$. It is therefore the q -zeta polynomial, and the same formula is valid at every integer m , including the negative ones. $\square$

Remark 9.5 (Two constructions at negative arguments). Proposition 9.4 and Lemma 3.1 are two different constructions of the q -zeta polynomial at negative arguments, and neither implies the other. The endpoint formula is uniform in m and is what makes every $[-r]_q$ reachable, through Lemma 9.6 below; it is special to coordinate downsets. The matrix continuation $e_0^\top A^{m-1} \mathbf{j}$, with its rank-filtration diagonalization, applies to an arbitrary finite poset with least element and an arbitrary normalized height, and is what makes Theorem 1.3 reachable. Both are kept.

Lemma 9.6 (The local negative kernel). *For every $r \geq 1$ and $a \geq 0$,*

$$q^a \left[\begin{matrix} a - r - 2 \\ a \end{matrix} \right]_q = [z^a] \prod_{j=0}^r (1 - q^{-j} z). \quad (9.8)$$

Proof. When $a \geq r + 2$, the product (9.6) with $m = -r$ contains a factor $1 - q^0$, so both sides vanish. If $0 \leq a \leq r + 1$, rewrite the negative powers by $1 - q^{-k} = -q^{-k}(1 - q^k)$ to get

$$q^a \begin{bmatrix} a - r - 2 \\ a \end{bmatrix}_q = (-1)^a q^{-ra + \binom{a}{2}} \begin{bmatrix} r + 1 \\ a \end{bmatrix}_q,$$

and expanding $\prod_{j=0}^r (1 - q^{-j}z)$ by elementary symmetric functions gives the same expression. This finite q -binomial identity also follows by induction on r , multiplying by $1 - q^{-r}z$. $\square$

Proof B of (1.3). By Proposition 9.4 and Lemma 9.6 with $r = 1$, the left side of (1.3) is the left side of (9.3) with every $f_i(z) = f(z) = (1 - z)(1 - q^{-1}z)$. Its transfer is $\mathcal{T}f(z) = 1 + q^{-1}z/(1 - z)$, whose coefficient is one in degree zero and q^{-1} in every positive degree. Hence Theorem 9.2 gives

$$\mathcal{Z}_q(P_\tau, [-1]_q) = (-1)^n \sum_{b \in \text{Max}(P_\tau)} \prod_{i: b_i > 0} q^{-1} = (-1)^n \sum_{b \in \text{Max}(P_\tau)} q^{-|\text{supp}(b)|},$$

and Lemma 2.3 identifies that statistic with the cov of the conjecture. No step of Section 3 was used. $\square$

9.5 A shorter derivation once capacity reciprocity is known

The decisive interpolation can alternatively be compressed to the two capacity values $u_i \in \{1, 2\}$. Put $t = q^{-1}$. For every $a \geq 0$,

$$[z^a](1 - z)(1 - tz) = (1 - t) \binom{a - 2}{a} + t \binom{a - 3}{a},$$

so multilinearity lets one combine (9.1) over all choices $u_i \in \{1, 2\}$. On the maximal-layer side the corresponding local factor at b becomes

$$(1 - t) \binom{b - 1}{b} + t \binom{b}{b} = \begin{cases} 1, & b = 0, \\ t, & b > 0, \end{cases}$$

which proves the conjecture directly. As Remark 7.2 records, this two-capacity collapse is the same content as the two-value interpolation table of Lemma 7.1, in binomial rather than z -polynomial language; the two were reached independently, from opposite ends of the cluster of arguments merged here. The general transfer theorem is kept nevertheless, because it is what exposes why the same method works at every negative q -integer and not only at $[-1]_q$.

10 All negative evaluations: positivity and combinatorics

Set $t = q^{-1}$ and $F_r(z) = \prod_{j=0}^r (1 - t^j z)$. A direct substitution into (9.2) gives

$$\mathcal{T}F_r(z) = \frac{\prod_{j=1}^r (1 - (1 - t^j)z)}{(1 - z)^r} = \prod_{j=1}^r \left(1 + \frac{t^j z}{1 - z} \right). \quad (10.1)$$

To select a coefficient of z^a , choose the s factors in which the nonconstant summand is used; their contribution is $e_s(t, t^2, \dots, t^r) z^s (1 - z)^{-s}$. For $a \geq 1$ this gives

$$[z^a] \mathcal{T}F_r(z) = \sum_{s=1}^{\min(r, a)} \binom{a - 1}{s - 1} e_s(t, t^2, \dots, t^r) = C_{r, a}(t),$$

the degree-zero coefficient being one, in agreement with (1.10). One may also use the compact Gaussian form

$$C_{r, a}(t) = \sum_{s=1}^{\min(r, a)} \binom{a - 1}{s - 1} t^{s(s+1)/2} \begin{bmatrix} r \\ s \end{bmatrix}_t \quad (a \geq 1). \quad (10.2)$$

Proof of the identity and positivity in Theorem 1.2. Proposition 9.4 and Lemma 9.6 express $\mathcal{Z}_q(P_\tau, [-r]_q)$ as the left side of (9.3) with every $f_i = F_r$. Equation (10.1) identifies the transferred coefficients as $C_{r,a}(t)$, which proves (1.11). Positivity and integrality are immediate from (1.10). The extreme-degree statements are proved in Proposition 10.2 below. $\square$

Nothing in Proof A goes past $[-1]_q$; Theorem 1.2 is the single largest piece of mathematics in this document that has only one proof.

10.1 A colored-composition interpretation

For $a > 0$, an object counted by $C_{r,a}(t)$ consists of a nonempty set of colors $1 \leq j_1 < \dots < j_s \leq r$ together with a composition $a = a_1 + \dots + a_s$ into s positive parts; its weight is $t^{j_1 + \dots + j_s}$. The colors are written in increasing order, so no factor $s!$ occurs, and there are $\binom{a-1}{s-1}$ choices of composition. For $a = 0$ there is one empty object of weight one.

Consequently the signed negative evaluation counts pairs consisting of a maximal lattice point b and, independently at each coordinate, one such colored composition of b_i . This explains the nonnegative coefficients without any appeal to cancellation in the original expression for the q -zeta polynomial. For instance,

$$\begin{aligned} C_{1,a}(t) &= t & (a \geq 1), \\ C_{2,a}(t) &= t + t^2 + (a-1)t^3 & (a \geq 1), \\ C_{3,a}(t) &= (t + t^2 + t^3) + (a-1)(t^3 + t^4 + t^5) + \binom{a-1}{2}t^6 & (a \geq 1), \end{aligned}$$

the usual convention making the last binomial zero when $a = 1, 2$.

10.2 The ordinary specialization at every r

Substituting $t = 1$ in (10.1) gives $(1 - z)^{-r}$, so

$$C_{r,a}(1) = \binom{r+a-1}{a}. \quad (10.3)$$

After the substitution $q^m = 1 + (q-1)X$, the j th local quotient in (9.6) tends coefficientwise to $(X + j - 2)/j$ as $q \rightarrow 1$, so the coefficientwise limit in the endpoint formula identifies the ordinary zeta polynomial and yields

$$(-1)^n \mathcal{Z}(P_\tau, -r) = \sum_{b \in \text{Max}(P_\tau)} \prod_{i \in E} \binom{r + b_i - 1}{b_i}. \quad (10.4)$$

At $r = 1$ this is the unweighted identity already covered by [9] and discussed in Section 7.2; it is not claimed as new. At general r it also follows directly by specializing all $u_i = r + 1$ in (9.1). The negative ordinary zeta values at $r > 1$ appear nowhere else in this document.

10.3 Sharp extreme degrees

Lemma 10.1 (Minimum support size). *The smallest support size of an element of $\text{Max}(P_\tau)$ is the number c_τ of maximal equivalence classes of τ .*

Proof. A maximal equivalence class F is a filter, so its complement is an ideal and every $b \in \text{Max}(P_\tau)$ satisfies $b(F) = n - b(E \setminus F) \geq |F| > 0$. The distinct maximal classes are disjoint, forcing at least one support coordinate in each.

For the converse, choose a representative in each maximal class. For every $i \in E$ choose a maximal class above the class of i and send one unit from i to the chosen representative there; let b_j count the units received by j . Every unit assigned to a member of an ideal I originates in I , since its source lies below its destination, so $b(I) \leq |I|$, and $|b| = n$; thus $b \in \text{Max}(P_\tau)$. Its support is exactly the set of chosen representatives, because every maximal class receives at least the units from its own members. $\square$

Proposition 10.2 (Sharp extreme degrees). *For $n, r \geq 1$ let $H_{\tau,r}(t) = \sum_{b \in \text{Max}(P_\tau)} \prod_i C_{r,b_i}(t)$. Then $H_{\tau,r}$ has degree exactly rn and leading coefficient one, its smallest exponent is c_τ , and the coefficient of t^{c_τ} is the number of maximal points of support size c_τ .*

Proof. For every $a > 0$ the smallest exponent in $C_{r,a}$ is one with coefficient one, from the single color 1, and positivity prevents cancellation; so the smallest exponent in the summand indexed by b is $|\text{supp}(b)|$, and Lemma 10.1 proves the assertion at the lower end.

A choice of $s \leq a$ distinct colors has weight exponent at most rs , hence at most ra . Equality with ra is possible only when $a = 1$: if $s < a$ the first bound is strictly smaller than ra , and if $s = a \geq 2$ then distinctness of the colors makes their sum strictly smaller than ra . Thus $\deg C_{r,a} \leq ra$, with strict inequality for $a \geq 2$, while $C_{r,1} = t + \cdots + t^r$ is monic of degree r . Every maximal point has total n , and the only one all of whose coordinates are at most one is $\mathbf{1}$; its summand $(t + \cdots + t^r)^n$ is monic of degree rn , and every other summand has smaller degree. $\square$

This completes the proof of Theorem 1.2.

11 Consequences: matroids, feasible supports, and coefficients

11.1 Two matroid invariants for the same coefficient

Fix $B \subseteq E$. Make a bipartite graph whose sources are $E \setminus B$ and whose destinations are E , with an edge $i \rightarrow j$ whenever $i \leq_\tau j$. Let M_B be the transversal matroid on the destination set E : a set S is independent when its elements can be matched to distinct sources. The matching exchange property follows by taking alternating paths in the union of two matchings, so these sets do form a matroid.

Proposition 11.1 (Tutte evaluation). *The independent sets of M_B are exactly*

$$\mathcal{I}_B = \{S \subseteq E : \mathbf{1}_B + \mathbf{1}_S \in P_\tau\}, \quad (11.1)$$

and $\text{rank } M_B = n - |B|$. If $T_M(x, y)$ denotes the Tutte polynomial, then

$$[\mathbf{z}^B] H_\tau(\mathbf{z}) = T_{M_B}(0, 1). \quad (11.2)$$

Proof. Membership in (11.1) is the family of inequalities $|S \cap I| \leq |I \setminus B|$ over ideals I . A matching implies them, because every destination in I must be matched from $I \setminus B$. Conversely, for $A \subseteq S$ apply the inequality to $D_\tau(A)$:

$$|A| \leq |S \cap D_\tau(A)| \leq |D_\tau(A) \setminus B|,$$

which are Hall's conditions for matching S into the available sources. There are $n - |B|$ sources, and the destination set $E \setminus B$ can be matched identically to them, which gives the rank.

By the subset expansion $T_M(x, y) = \sum_{A \subseteq E} (x - 1)^{\text{rank } M - \text{rank}(A)} (y - 1)^{|A| - \text{rank}(A)}$, at $(x, y) = (0, 1)$ only independent sets survive, giving $T_M(0, 1) = (-1)^{\text{rank } M} \sum_{A \text{ independent}} (-1)^{|A|}$. Substitute $\text{rank } M_B = n - |B|$ and use (1.9). $\square$

The same coefficient carries a second, different standard invariant. By Remark 8.7 the family $\mathcal{I}_B$ is the independent-set family $\mathcal{I}_{H_B}$ of Engine 2, a transversal matroid of rank $n - |B|$ whose private elements form a basis. Define the h -polynomial of its independence complex directly by

$$h_B(t) = \sum_{T \in \mathcal{I}_B} t^{|T|} (1-t)^{n-|B|-|T|}. \quad (11.3)$$

Its top coefficient is $(-1)^{n-|B|} \sum_T (-1)^{|T|} = N_\tau(B)$. So (11.2) and (11.3) are two standard matroid invariants for one and the same number, reached from two different literatures; (11.3) needs no matroid theorem at all, while (11.2) places the coefficient inside the Tutte formalism. Both are recorded for that reason.

In either reading, the signs in the Boolean sum encode a familiar matroid evaluation, but the equality with a count of maximal preorder points still rests on Section 6 or Section 8. Nonnegativity alone would not determine that count.

11.2 Exactly which supports occur

Call $B \subseteq E$ *cofinal* if every $r \in E$ satisfies $r \leq_\tau s$ for some $s \in B$; equivalently, B meets every maximal equivalence class of τ , that is, every maximal element of the quotient partial order. Two statements are recorded, a qualitative one and a quantitative one; neither follows from the other.

Proposition 11.2 (Support feasibility, qualitative form). *For every $B \subseteq E$, $N_\tau(B) > 0$ if and only if B is cofinal. Consequently $\eta_\tau(B) = 0$ on noncofinal supports, and on cofinal supports its sign is exactly $(-1)^{n-|B|}$.*

Proof. Let C be a maximal class; its complement is an ideal, and a maximal point has total n , so $a(E \setminus C) \leq n - |C|$ forces $a(C) \geq |C| > 0$, and the support of a meets C .

Conversely suppose B meets every maximal class. Retain the unit of each source $b \in B$ at its own destination; for each source $i \notin B$ choose a maximal class above the class of i and some $b \in B$ in that class, and send the source's unit to b . This is allowed transport and uses every source, so its destination vector has total n and exact support B , hence is maximal. The same conclusion follows from Lemma 8.6: positivity means that every resource in $E \setminus B$ can choose some neighbour in B , the choices not needing to be distinct because the degree vector records multiplicities. The assertion about signs is Theorem 1.1. $\square$

The quantitative counterpart is Lemma 10.1 together with Proposition 10.2: the *minimum* support size is the number c_τ of maximal equivalence classes, and the coefficient at that exponent counts the maximal points of that support size. Its constructive unit-routing proof is kept because Proposition 10.2, and hence the degree statement of Theorem 1.2, depends on it.

It follows that the coefficient of u^k in $H_\tau(u)$ is positive exactly for $c_\tau \leq k \leq n$, with no internal zero coefficients. This does *not* assert log-concavity, real-rootedness, or any stronger coefficient property.

11.3 Coefficients of the support polynomial

Proposition 11.3 (Extreme and near-extreme coefficients). *For $n \geq 1$,*

$$[u^n]H_\tau(u) = 1, \quad [u^{n-1}]H_\tau(u) = \#\{(i, j) \in E^2 : i \neq j, i \leq_\tau j\} = \sum_{i \in E} (|U_\tau(i)| - 1). \quad (11.4)$$

If the quotient preorder has exactly one maximal class C , then $[u]H_\tau(u) = |C|$; if it has more than one maximal class, this coefficient is zero. Equivalently, $[u]H_\tau(u) = \#\{s \in E : r \leq_\tau s \text{ for all } r \in$

$E\}$. For $n \geq 2$, writing $S_{ru} = E \setminus \{r, u\}$,

$$[u^{n-2}]H_\tau(u) = \sum_{\{r,u\} \subseteq E} \left(|U_\tau(r) \cap S_{ru}| |U_\tau(u) \cap S_{ru}| - \binom{|U_\tau(r) \cap U_\tau(u) \cap S_{ru}|}{2} \right). \quad (11.5)$$

Proof. The only positive-coordinate integer vector of total n is $\mathbf{1}$. A vector of total n with support of size $n - 1$ is uniquely $\mathbf{1} - e_i + e_j$ with $i \neq j$, and it satisfies the ideal inequalities exactly when every ideal containing j also contains i , which is equivalent to $i \leq_\tau j$; for necessity take the ideal $D_\tau(j)$. For singleton support the vector must be ne_j , and the transport criterion says that every source lies below j , which in a finite preorder happens exactly when there is a unique maximal class and j lies in it.

For (11.5), use Lemma 8.6: fix the two residual resources r, u , and let their target sets inside S_{ru} have sizes a, b with intersection of size d . There are ab ordered choices. Two distinct choices produce the same degree vector exactly when two distinct common targets are exchanged between r and u , so each of the $\binom{d}{2}$ unordered pairs of common targets creates exactly one excess count, and no degree vector has any other multiplicity. The number of distinct vectors is therefore $ab - \binom{d}{2}$. Summing over residual pairs proves the formula. $\square$

Formula (11.5) is the only third-from-the-top coefficient formula available here, and the product rule below is the only statement about disjoint unions; they came from different drafts and neither subsumes the other. In particular, the negative q -evaluation has pole order exactly n at $q = 0$:

$$(-1)^n \mathcal{Z}_q(P_\tau, -q^{-1}) = q^{-n} + c'_\tau q^{-(n-1)} + \text{higher powers of } q,$$

where c'_τ counts the ordered nonreflexive comparisons in (11.4). Equivalent but distinct elements contribute in both directions.

11.4 Products and an inclusion–exclusion algorithm

If $\tau = \tau_1 \sqcup \tau_2$ is a disjoint union on disjoint ground sets, then $P_\tau = P_{\tau_1} \times P_{\tau_2}$, its maximal points are products of maximal points, and therefore

$$H_\tau(\mathbf{z}) = H_{\tau_1}(\mathbf{z}_{E_1}) H_{\tau_2}(\mathbf{z}_{E_2}),$$

with no additional interaction between components.

The cumulative counts can be inverted on the Boolean lattice:

$$[\mathbf{z}^B]H_\tau(\mathbf{z}) = \sum_{T \subseteq B} (-1)^{|B|-|T|} B_\tau(T) = (-1)^n \sum_{T \subseteq B} (-1)^{|B|-|T|} L_{\tau^*}(-\mathbf{1} - \mathbf{1}_T). \quad (11.6)$$

Once the 2^n cumulative values are known, all exact-support counts follow in $O(n2^n)$ additions by the Boolean Möbius transform. This is an exact finite procedure, not a claim that obtaining the cumulative counts is polynomial-time.

11.5 Arbitrary heights and a multivariate matrix identity

Theorem 1.3 was proved in Section 7. Two consequences deserve separate statements.

Corollary 11.4 (Linear weights). *For positive integers w_i and $h(a) = \sum_i w_i a_i$,*

$$\mathcal{Z}_{q,h}(P_\tau, [-1]_q) = (-1)^n \sum_{b \in \text{Max}(P_\tau)} q^{-\sum_{i \in \text{supp}(b)} w_i}. \quad (11.7)$$

Proof. Substitute $h(\mathbf{1}_{\text{supp}(b)}) = \sum_{i \in \text{supp}(b)} w_i$ in (1.12). The exponent involves the Boolean support vector and not $h(b)$; the Möbius function in (3.5) is exactly what enforces that distinction. $\square$

Corollary 11.5 (Multivariate matrix identity). *Let z_i be commuting indeterminates, put $\mathbf{z}^a = \prod_i z_i^{a_i}$ and $D(\mathbf{z}) = \text{diag}(\mathbf{z}^a)_{a \in P_\tau}$. Then, over $\mathbb{Q}(z_i : i \in E)$,*

$$e_0^\top (ZD(\mathbf{z}))^{-2} \mathbf{j} = (-1)^n \sum_{b \in \text{Max}(P_\tau)} \prod_{i \in \text{supp}(b)} z_i^{-1}. \quad (11.8)$$

Proof. The purely algebraic equality $e_0^\top (ZD)^{-2} \mathbf{j} = e_0^\top \mu D^{-1} \mu \mathbf{j}$ still holds, and (3.4) reduces it to $\sum_B (-1)^{|B|} \eta_\tau(B) \prod_{i \in B} z_i^{-1}$. Apply Theorem 1.1. $\square$

Equation (11.8) is a multivariate *matrix* identity; it is not a claim that some multivariate interpolation polynomial has been constructed, and no such object is needed. Setting all $z_i = 1$ is legitimate because Z is invertible, and returns $\mathcal{Z}(P_\tau, -1) = (-1)^n \# \text{Max}(P_\tau)$ as in Section 7.2.

12 Explicit examples

12.1 Antichains, discrete preorders, and a single equivalence class

For an n -element antichain the defining inequalities reduce to $0 \leq x_i \leq 1$, so $P_\tau = \{0, 1\}^E$ and the only maximal point is $\mathbf{1}$:

$$H_\tau(\mathbf{z}) = \prod_i z_i, \quad \mathcal{Z}_q(P_\tau, -q^{-1}) = (-1)^n q^{-n}, \quad \mathcal{Z}_q(P_\tau, X) = X^n,$$

the last because a multichain is determined coordinatewise by its number of ones, which may be $0, \dots, m-1$. At the other negative arguments, $(-1)^n \mathcal{Z}_q(P_\tau, [-r]_q) = (t + t^2 + \dots + t^r)^n$. This example checks both the sign and the shift in the length of the multichains.

At the opposite extreme let all n elements be equivalent. The only nonempty ideal is E , so $P_\tau = \{a \geq 0 : |a| \leq n\}$ and the maximal points are the weak compositions of n into n parts. For a prescribed support B of size $b \geq 1$ the positive coordinates form a composition of n into b positive parts, so, by placing $b-1$ separators in the $n-1$ internal gaps,

$$[\mathbf{z}^B] H_\tau(\mathbf{z}) = \binom{n-1}{b-1}, \quad H_\tau(u) = \sum_{b=1}^n \binom{n}{b} \binom{n-1}{b-1} u^b. \quad (12.1)$$

For $n=3$ this is $3u+6u^2+u^3$ with 10 maximal points, and for $n=5$ it is $5u+40u^2+60u^3+20u^4+u^5$ with 126. The example also shows that an equivalence class must not be collapsed to one coordinate: its cardinality affects both the polytope and the support statistic.

12.2 Chains

For the partial order $1 < 2 < \dots < n$ the maximal vectors are described by $a_1 + \dots + a_i \leq i$ for $1 \leq i < n$ together with $|a| = n$. For $n=3$ they are

$$(0, 0, 3), \quad (0, 1, 2), \quad (0, 2, 1), \quad (1, 0, 2), \quad (1, 1, 1),$$

so $H_\tau(u) = u + 3u^2 + u^3$, with full multivariate form $z_3 + 2z_2z_3 + z_1z_3 + z_1z_2z_3$. The nonzero supportwise counts and their signed Boolean sums are

B	$N_\tau(B)$	$\eta_\tau(B)$
$\{3\}$	1	1
$\{1, 3\}$	1	-1
$\{2, 3\}$	2	-2
$\{1, 2, 3\}$	1	1

All other supports have both counts zero, because they are not cofinal. For $B = \{2, 3\}$ the feasible S are $\emptyset, \{1\}, \{2\}, \{3\}$, with alternating sum -2 ; for $B = \{3\}$ the independent-subset counts by size in the private residual graph are $1, 3, 3$, with alternating sum 1 . Hence $\mathcal{Z}_q(P_\tau, -q^{-1}) = -q^{-1} - 3q^{-2} - q^{-3}$, and at the next negative argument

$$(-1)^3 \mathcal{Z}_q(P_\tau, [-2]_q) = t + 4t^2 + 9t^3 + 9t^4 + 6t^5 + t^6.$$

For the nonlinear height $h(a) = |a|^2$, Theorem 1.3 instead gives

$$\mathcal{Z}_{q,h}(P_\tau, [-1]_q) = -q^{-1} - 3q^{-4} - q^{-9},$$

which is $-353/512$ at $q = 2$, against $-11/8$ in the linear case. The verifier obtains both by interpolation from positive multichain values, with a degree bound of 9 in the nonlinear case, without ever using a negative-evaluation formula.

For $n = 5$ direct enumeration gives 132 lattice points, 42 maximal vectors,

$$H_\tau(u) = u + 10u^2 + 20u^3 + 10u^4 + u^5, \quad \mathcal{Z}_2(P_\tau, -1/2) = -\frac{197}{32}.$$

This example has a unique maximum, so its least support size is one, and its coefficient at u^4 is the ten ordered comparisons $i < j$.

12.3 A fork, and why the direction of the preorder matters

For V : $1 < 2$ and $1 < 3$, the three maximal points are $(0, 1, 2)$, $(0, 2, 1)$ and $(1, 1, 1)$, so

$$H_V(\mathbf{z}) = 2z_2z_3 + z_1z_2z_3, \quad \mathcal{Z}_q(P_V, -q^{-1}) = -2q^{-2} - q^{-3}, \quad H_V(t) = 2t^2 + t^3.$$

There are two maximal classes, which explains the absence of a t term before any computation. This small example already distinguishes maximal-point support from rank: all three points have rank three, but their support sizes are two, two and three, so replacing the cover statistic by the rank would give an incorrect formula.

For the dual preorder V^* , relabelled so that 3 lies above the incomparable 1, 2, one gets instead $H_{V^*}(t) = t + 2t^2 + t^3$. Both lattice-point posets have 12 elements, but their numbers of maxima are 3 and 4. Neither the maximal-support polynomial nor its q -zeta specialization should therefore be confused with the different support-enumerator duality studied in [8].

12.4 A diamond

For the partial order $1 < 2, 1 < 3, 2 < 4, 3 < 4$ one finds $|P_\tau| = 38$, $|\text{Max}(P_\tau)| = 12$ and

$$H_\tau(t) = t + 5t^2 + 5t^3 + t^4.$$

This is the only four-element example worked out here.

12.5 The five-element example in the source

Take $E = \{a, b, c, d, e\}$ with $a \sim c$, with $a, c < e$, with $b < d$ and $b < e$, and with no other comparisons except those forced by reflexivity and transitivity. This is the running example of [1, Sections 3.1 and 4.4]. Besides nonnegativity, its polytope is described by

$$\begin{aligned} x_a + x_c &\leq 2, & x_b &\leq 1, \\ x_a + x_b + x_c + x_e &\leq 4, & x_b + x_d &\leq 2, \\ x_a + x_b + x_c + x_d + x_e &\leq 5. \end{aligned}$$

There are 92 lattice points and 18 maximal ones. The maximal classes are $\{d\}$ and $\{e\}$, so every support contains both d and e . The complete exact-support distribution is

Support	Count	Support	Count
$\{d, e\}$	2	$\{a, b, d, e\}$	2
$\{a, d, e\}$	4	$\{a, c, d, e\}$	2
$\{b, d, e\}$	1	$\{b, c, d, e\}$	2
$\{c, d, e\}$	4	$\{a, b, c, d, e\}$	1

Thus

$$H_\tau(\mathbf{z}) = z_d z_e (2 + 4z_a + z_b + 4z_c + 2z_a z_b + 2z_a z_c + 2z_b z_c + z_a z_b z_c), \quad (12.2)$$

$$H_\tau(u) = 2u^2 + 9u^3 + 6u^4 + u^5, \quad (12.3)$$

and Theorem 1.1 gives

$$\mathcal{Z}_q(P_\tau, -q^{-1}) = -\frac{2q^3 + 9q^2 + 6q + 1}{q^5} = -\frac{(2q+1)(q^2+4q+1)}{q^5},$$

which agrees with the expression printed on page 15 of [1]. The support table is a refinement of that expression, not merely its total value -18 at $q = 1$; (12.2) is finer still, and is the multivariate refinement of the source's own example. At the next negative argument, Theorem 1.2 gives

$$\begin{aligned} (-1)^5 \mathcal{Z}_q(P_\tau, [-2]_q) &= 2t^2 + 13t^3 + 41t^4 + 76t^5 + 94t^6 \\ &\quad + 76t^7 + 40t^8 + 11t^9 + t^{10}, \end{aligned}$$

whose lowest exponent 2 agrees with the two maximal equivalence classes, as Proposition 10.2 requires.

13 Why the hypotheses cannot simply be discarded

Three examples are recorded, carrying four distinct failure statements between them. Each bounds a different hypothesis, and none of them can be deduced from the others: the first shows that purity of the downset is not enough, the second that transitivity is not implied by the rest of the transport picture, and the third that unit capacities are essential.

13.1 Purity alone is insufficient: a pure downset, two failures

It is natural to ask whether (1.3), or the transfer identity (9.3) behind Proof B, holds for every finite pure coordinate downset in $\mathbb{N}^E$ containing the Boolean cube. Neither does. Let

$$P = [\mathbf{0}, (1, 1, 1)] \cup [\mathbf{0}, (3, 0, 0)] \subseteq \mathbb{N}^3, \quad (13.1)$$

Preorder	$ P_\tau $	$ \text{Max}(P_\tau) $	$H_\tau(t)$	$\mathcal{Z}_q(P_\tau, -q^{-1}) _{q=2}$
Antichain, $n = 3$	8	1	t^3	$-1/8$
Chain, $n = 3$	14	5	$t + 3t^2 + t^3$	$-11/8$
V (fork), $n = 3$	12	3	$2t^2 + t^3$	$-5/8$
V^* , $n = 3$	12	4	$t + 2t^2 + t^3$	$-9/8$
One equivalence class, $n = 3$	20	10	$3t + 6t^2 + t^3$	$-25/8$
Diamond, $n = 4$	38	12	$t + 5t^2 + 5t^3 + t^4$	—
Chain, $n = 5$	132	42	$t + 10t^2 + 20t^3 + 10t^4 + t^5$	$-197/32$
One equivalence class, $n = 5$	252	126	$5t + 40t^2 + 60t^3 + 20t^4 + t^5$	$-681/32$
Source example, $n = 5$	92	18	$2t^2 + 9t^3 + 6t^4 + t^5$	$-65/32$

Table 1: Exact examples, all computed independently by the included verifiers. The diamond has relations $1 < 2$, $1 < 3$, $2 < 4$, $3 < 4$. The last column is the negative q -evaluation at $q = 2$; the seven rows for which it is given are also recorded in `data/examples.csv`.

a finite graded coordinate downset containing every Boolean vector, whose two maximal elements $(1, 1, 1)$ and $(3, 0, 0)$ both have rank three.

Failure of the supportwise identity and of the q -zeta value. For $B = \{1\}$ the terms with $S \subseteq \{2, 3\}$ cancel, while $S = \{1\}$ contributes -1 ; so $\eta_P(\{1\}) = -1$, whereas $(-1)^{3-|B|}N_P(\{1\}) = +1$. The only other nonvanishing η_P is at $B = \{1, 2, 3\}$, where it is 1, so (3.5) gives

$$\mathcal{Z}_q(P, -q^{-1}) = q^{-1} - q^{-3},$$

whereas the proposed extension would give $-q^{-1} - q^{-3}$. At $q = 2$ the discrepancy is $3/8$ against $-5/8$.

Failure of the transfer identity. With $f_1(z) = z^2$ and $f_2 = f_3 = 1$, the two sides of (9.3) are 1 and -1 . This is immediate from $\mathcal{T}(z^2) = z^2/(1-z)$ and $\mathcal{T}(1) = 1-z$: the left side counts the single point $(2, 0, 0) \in P$, while on the right only $b = (3, 0, 0)$ contributes, with weight 1 and sign $(-1)^3$.

Both statements are kept. They are genuinely different: the first says the conclusion of Theorem 1.1 fails for P , the second says the much more general machine of Theorem 9.2 fails for P , so capacity reciprocity is a real restriction inside Proof B and not an artifact of its presentation. This counterexample locates a genuine restriction of the theorem: its proofs use the preorder allocation structure, not merely purity and product-of-chains intervals.

13.2 A reflexive, nontransitive incidence system

One could instead try to prove the theorem for any downset generated by assigning unit sources to allowed destinations, with each source allowed to stay in place. The following example rules that out, and it is not covered by Section 13.1, because here the transport representation is present and only transitivity is missing.

On three coordinates give the sources the neighbourhoods

$$A_1 = \{1, 2\}, \quad A_2 = \{2, 3\}, \quad A_3 = \{1, 3\}.$$

The relation is reflexive but not transitive. Its distinct full-assignment vectors are the six permutations of $(2, 1, 0)$ together with $(1, 1, 1)$. Let P be their coordinate downset. All seven generators are maximal, and

$$H_P(\mathbf{z}) = 2(z_1z_2 + z_1z_3 + z_2z_3) + z_1z_2z_3. \quad (13.2)$$

The downset contains every vector of total at most two, and among total-three vectors exactly the seven just listed. The general Boolean localization formula (3.6) gives

$$-R_P(\mathbf{z}) = z_1 + z_2 + z_3 + 2(z_1z_2 + z_1z_3 + z_2z_3) + z_1z_2z_3. \quad (13.3)$$

For instance, at the coefficient of z_1 the feasible S are all the proper subsets of $\{1, 2, 3\}$, whose alternating sum is $1 - 3 + 3 = 1$; yet there is no maximal point supported only at coordinate one. In diagonal form the true maximal-support polynomial is $6u^2 + u^3$, whereas the signed negative q -zeta value is $3u + 6u^2 + u^3$.

Reflexivity, purity, unit supplies, coordinate-box intervals and a transport representation are therefore jointly insufficient. Transitivity enters decisively when arbitrary neighbourhood constraints are replaced by ideal or filter inequalities, which is exactly what Lemma 8.5 and the filter identification inside Proposition 5.1 exploit.

13.3 Arbitrary positive capacities do not satisfy the same formula

The auxiliary polynomial $L_{\tau^*}(\mathbf{t})$ has independent capacity variables, but this does not extend the main q -formula unchanged to $Q_{\tau}(\mathbf{c})$ with arbitrary capacities. Take one coordinate of capacity two: the lattice-point poset is the chain $0 < 1 < 2$, and (3.7) gives

$$\mathcal{Z}_q(P, -q^{-1}) = 1 - (1 + q^{-1}) + q^{-1} = 0.$$

Its sole maximal point has support size one, so neither a sign based on ambient dimension nor one based on maximal rank can produce the asserted support formula. The unit-capacity model in the main theorem is essential.

13.4 What the result does not settle

This document does not establish the other real-rootedness, magic-positivity or γ -positivity conjectures for preorder polytopes, and nonnegativity of the coefficients in (1.11) does not imply real-rootedness or log-concavity; no such inference is made. As recorded in Section 1.5, H_{τ} counts only maximal points, whereas the preorder h -polynomial of [1, Section 5] counts support sizes over all lattice points; these are different polynomials.

We also do not provide a canonical pointwise bijection between $B_{\tau}(T)$ and $\Lambda_{\tau}(T)$: in Engine 1 that equality uses the imported bipartite duality, and a triangulation-dependent bijection could be sought through that theorem but is not needed. Similarly, no direct sign-reversing involution realizing (1.9) is constructed. In Proof B the operator $\mathcal{T}$ is an algebraic involution on local series, not a combinatorial matching of individual global summands; the colored-composition model of Section 10 gives a positive interpretation of the final answer without pretending to supply such a matching.

Two open problems are recorded rather than solved. First, characterize the finite coordinate downsets for which the transfer identity (9.3) holds for every family of local series; the example of Section 13.1 shows the class is strictly smaller than the pure downsets containing the Boolean cube. Second, characterize the finite downsets satisfying the supportwise identity (1.9). Neither is claimed as an additional solved problem here.

14 Exact computation and reproducibility

Three independent verifiers are shipped, one from each of the merged packages. They test three different theorems and none subsumes the others, so all three are retained rather than unified:

- `code/verify.py` — the support/interior/transport suite, with four random-construction modes and the weighted lattice polynomial at both nonnegative and negative capacities;
- `code/verify_transfer.py` — the $[-r]_q$ polynomial identities at $r = 1, \dots, 4$, arbitrary signed local-kernel transfer tests, direct capacity and reciprocity counts, and multichain dynamic-programming interpolation;

- `code/verify_supportwise.py` — the inverse-matrix evaluations, the nonlinear-height interpolation, and the standalone bipartite-lemma suite;
- `code/test_unit.py` — ten fast normalization and boundary tests.

All of them use only the Python standard library, all counts use integers, and all interpolations and polynomial evaluations use `fractions.Fraction`. There is no floating-point tolerance, computer algebra service or hidden dataset; the only floating-point quantity recorded anywhere is elapsed wall-clock time. A failed equality raises an exception instead of silently recording approximate agreement.

Note that the runs used *different seeds* and *different sampling depths*, and this has been preserved deliberately: `verify.py` and `verify_transfer.py` use seed 20260920, while `verify_supportwise.py` uses seed 26092049; and at sizes 6, 7, 8 the first samples 16 relations per size while the third samples 64. The sampled populations are therefore genuinely different, and that independence is part of the evidentiary value.

14.1 Exhaustive enumeration

For n labelled elements each program considers every reflexive relation by choosing, independently, the $n - 1$ possible off-diagonal entries of each row, and then accepts exactly the transitive ones: whenever row i contains j , row j must be a subset of row i . There are $2^{n(n-1)}$ candidates and each labelled preorder is visited exactly once. No antisymmetry condition is imposed, so preorders with genuine equivalence classes are included, and isomorphism classes are not collapsed.

For every accepted relation `verify.py` independently computes the maximal candidates from the ideal inequalities at total n , their exact support counts by enumeration, and the alternating Boolean formula. It also counts every interior-capacity case $\mathbf{1} + \mathbf{1}_T$ directly from the strict filter inequalities, and compares the maximal candidates with all distinct full-assignment vectors in the transport model.

n	Labelled preorders	Support identities	Interior counts
0	1	1	1
1	1	2	2
2	4	16	16
3	29	232	232
4	355	5,680	5,680
5	6,942	222,144	222,144
Total	7,332	228,075	228,075

All comparisons passed. The counts in the two right columns include zero coefficients and empty support sets, not only successful nonempty supports, and the interior column is checked at full exhaustive depth. The assignment-model comparison was performed for each of the 7,332 preorders.

The supportwise verifier covers the same exhaustive range independently: 7,332 labelled preorders and 228,075 support identities, plus 64 distinct seeded preorders at each of sizes 6, 7, 8, contributing another 28,672 identities for a total of 256,747. Its sampled relations are saved as row bitmasks, so they can be recovered independently of the pseudorandom generator. The transfer verifier’s default exhaustive range is sizes $0, \dots, 4$, that is $1 + 1 + 4 + 29 + 355 = 390$ relations; it accepts `-max-n 5`.

14.2 Additional cases and independent routes

Using seed 20260920, `verify.py` generates sixteen cases at each of sizes six, seven and eight, alternating four construction modes: sparse acyclic relations, dense acyclic relations, closures of arbitrary directed relations, and partial orders on randomly chosen equivalence blocks. In the recorded run all sixteen relations at each size were distinct. This checks another 7,168 exact-support coefficients. The interior and assignment checks were also run on the size-six and size-seven cases; they were omitted at size eight in this test profile, which is stated openly rather than implied.

For each of the five named examples a separate routine counts *positive* weighted multichains at $m = 2, \dots, n + 2$, interpolates in the actual nodes $[m]_q$, evaluates at $-1/q$ and compares with the maximal-support polynomial, without invoking the Boolean negative-evaluation formula. At $q = 1, 2, 3, 5$ all twenty comparisons passed; for the source example the values were

$$-18, \quad -\frac{65}{32}, \quad -\frac{154}{243}, \quad -\frac{506}{3125}.$$

This independent route is particularly useful for checking the $m - 1$ chain length and the -2 matrix exponent. The weighted lattice polynomial was also checked in 156 evaluations on the five examples: twenty nonnegative capacity vectors against direct lattice counts, zero capacities included, and 136 vectors in $\{-1, -2\}^E$ against the support-interior identity. Finally, the nontransitive example of Section 13.2 is explicitly computed and checked to *fail* the extension it is meant to refute.

The transfer verifier checks, over its exhaustive range: every coefficient of the multivariate identity of Section 9.3; the full transfer (9.3) on deterministically generated signed integer local polynomials, which is a sample of weights and not an exhaustive check over all weights; and the negative evaluations as exact polynomials in t for $r = 1, 2, 3, 4$, the left side built from the finite product of Lemma 9.6 and the right side built independently from subsets of colors and compositions, with the leading and smallest-degree claims of Proposition 10.2 checked at the same time. To avoid merely comparing two rewritings of the same Gaussian expression, a further test computes positive-length multichain weights by the direct dynamic program

$$V_1(a) = q^{|a|}, \quad V_{\ell+1}(a) = q^{|a|} \sum_{b \leq a} V_{\ell}(b),$$

and reconstructs the polynomial by exact Lagrange interpolation at $q = 2, 3$, one further positive argument checking the interpolation before all four negative arguments are compared against (1.11). Independent-capacity counts were checked by direct lattice enumeration for all nonempty preorders of size at most three, all $u_i \in \{1, 2\}$ and $v \in \{0, 1, 2\}$, with reciprocity also checked at $v = 1, 2$.

Check family	Number passed
Exhaustively enumerated labelled preorders, transfer verifier	390
Full multivariate support identities in that range	390
Signed local-polynomial transfer tests in that range	390
Negative-evaluation polynomial identities, $r = 1, \dots, 4$	1,560
Direct capacity counts and reciprocity evaluations	1,250
Positive and negative interpolation comparisons	410
Additional deterministic random preorders, sizes 5–8	24

Table 2: Recorded output of `code/verify_transfer.py` at its default settings `-max-n 4`, `-max-r 4`, seed 20260920. The six worked examples also receive all structural and polynomial checks.

The supportwise verifier adds checks that exist nowhere else. For every preorder of size at most four, two triangular linear solves compute $e_0^T(ZD)^{-2}\mathbf{j}$ directly, at $q = 2$, $q = 3/2$ and with

independent coordinate weights $z_i = 2, 3, 5, 7$, for 1,170 matrix evaluations in total; seven named examples are checked by interpolating the q -zeta polynomial from positive multichain values at $q = 2$, and the three-element chain receives an eighth interpolation using the nonlinear height $h = \lfloor \cdot \rfloor^2$ with a degree bound of 9. Above all, Lemma 8.2 is tested *standalone*, in isolation from the preorder application, on all 5,058 labelled bipartite graphs with $0 \leq |R| \leq 3$ and $0 \leq |C| \leq 4$ and on 300 further seeded graphs of sizes $(4, 4)$, $(5, 4)$, $(4, 5)$, comparing three independently computed counts: independent subsets of the private-element graph, full-allocation degree vectors, and the integer vectors satisfying (8.7). All tests passed.

14.3 Two finite algorithms, and what they cost

Equation (1.9) gives a direct exact procedure for the whole maximal-support polynomial: enumerate $B, S \subseteq E$, test $\mathbf{1}_B + \mathbf{1}_S \in P_\tau$, and accumulate the signed sums. There are 4^n membership tests; using the allocation interpretation, each is a bipartite-matching test on at most $2n$ demand copies and n resources, and sums exceeding n can be rejected immediately. A basic augmenting-path implementation gives an $O(4^n n^3)$ arithmetic-operation upper bound, including all support counts. This is an exponential algorithm, not a claim of polynomial-time enumeration in n .

Alternatively, for a fixed support B , enumerate the distinct degree vectors of full allocations in H_B by dynamic programming: begin with the zero vector on B , and for each resource $r \in E \setminus B$ replace the current set D by

$$\bigcup_{s \in B: r \leq_\tau s} \{d + e_s : d \in D\}.$$

The cardinality of the final set is $N_\tau(B)$. The union is a *set* union: failing to deduplicate would count allocations rather than maximal vectors. The shipped verifiers use a third engineering choice suited to small exhaustive cases, encoding all ideals as a bitset and precomputing which subset inequalities each candidate vector violates; this changes neither the statement being checked nor its exact integer arithmetic.

If only the diagonal polynomial $H_\tau(u)$ is required, (3.7) uses 3^n vectors, each contributing $(-1)^{n+s}(1+u)^s u^d$, and a direct implementation takes $O(nj(\tau)3^n)$ elementary arithmetic operations, where $j(\tau)$ is the number of ideals, with no construction of a large incidence matrix. This bound ignores integer bit costs and is a transparent upper bound, not a sharp complexity result. Against it, enumeration of all rank- n candidate vectors requires at most $\binom{2n-1}{n-1}$ candidates when $n > 0$, which is often competitive. The value of the Boolean formula is therefore not a universal speed claim; it is structural locality, the matroid interpretation, and the ability to check two genuinely different computations against one another.

14.4 Running and reusing the programs

From the extracted archive directory, the recorded profiles are

```
python code/verify.py --exhaustive 5 --random-per-size 16 \
    --seed 20260920 --output data/verification.json
python code/verify_transfer.py --output data/verification_transfer.json
python code/verify_supportwise.py --output data/verification_supportwise.json
python -m unittest discover -s code -v
```

On PowerShell, put each command on one line, or replace the shell continuation with PowerShell's continuation character. A smaller smoke test is

```
python code/verify.py --exhaustive 3 --random-per-size 4 \
    --output data/smoke.json
```

Do not run Python with `-O`, since some test assertions use `assert`.

The row-mask representation is explicit and shared by all three programs: bit j in `rows[i]` means $i \leq_\tau j$, with zero-based labels. For example, $(21, 26, 21, 8, 16)$ encodes the five-element source example in the order a, b, c, d, e ; then `boolean_coefficients(rows)[24]` is 2, the exact-support count for $\{d, e\}$, and `direct_qzeta_negative(rows, 2)` is $-65/32$. Library functions expose ideal generation, lattice points, maximal points, exact support coefficients, cumulative support counts, direct interior counts, and positive-chain interpolation. Invalid relations should be checked with `is_preorder`; transitive closure is available as a separate, explicit operation, and a nontransitive relation must not be confused with its transitive closure, which is never a silently applied repair.

The JSON records the actual relations used in the additional tests and a SHA-256 digest of the deterministic exhaustive witness stream. That digest helps detect accidental changes to the run; it is not a mathematical certificate in the proof-assistant sense, and no checksum manifest of the archive is shipped or intended.

The source paper already reports verification of its original, univariate conjecture through size six [1, p. 15]. The role of these computations is independent reproducibility and testing of the stronger support-by-support identity, its two geometric engines, and the negative evaluations; it is not a claim to exceed that original size bound. Neither the original checks nor these establish an all-size theorem. That role belongs to the arguments in Sections 3–10.

15 Proof audit and research status

15.1 Dependencies

Two substantial results are imported: Postnikov’s lattice-count formula (Imported theorem 4.1) and Ehrhart–Macdonald reciprocity (4.6). Engine 1 of Proof A imports one more, Postnikov’s bipartite duality (Imported theorem 4.2); Engine 2 and Proof B do not. Hall’s theorem is used by Proof A and is proved in Appendix A; it is entirely absent from Proof B. Everything else is proved here: the Minkowski realization with independent capacities, the filter identification of draconian vectors, the polynomial extension of reciprocity, the incidence-matrix continuation and Boolean localization, the support–interior identity, the private-element Euler identity, the residual and privatization lemmas, the coefficient involution, the Gaussian endpoint formula, and the positive local factors. The filter identification is included with explicit credit to its appearance in [1, Lemma 4.1], and the overlaps with [9] are listed in Section 1.5 and flagged again at Remarks 5.3 and 5.5. No theorem asserted by another report in the supplied manifest is used as a proof source.

15.2 Normalization errors the proofs are sensitive to

The following points make the delicate conventions explicit. They merge the audits of the three source drafts; where two drafts recorded the same point differently, both formulations are kept.

1. **Lower covers and rank are different.** Covers are $a - e_i$ for positive coordinates; the exponent in the conjecture is the support size, not the common maximal rank n .
2. **The matrix is ZD , and the exponent is -2 .** Its column weighting gives the sum of the heights of the chain entries. There are $m - 1$ entries, so the negative evaluation uses A^{-2} , not A^{-1} . The independent multichain-interpolation tests were designed specifically to detect an error here.
3. **Repeated height eigenvalues cause no Jordan blocks.** The rank (or height) filtration proves annihilation by the *squarefree* polynomial $\prod_j (X - q^{g_j})$ over $\mathbb{Q}(q)$. The argument does not rely on upper triangularity alone.

4. **Boolean localization localizes only the middle vector.** The Möbius factor $\mu(\mathbf{0}, a)$ forces a to be Boolean, but $a + \mathbf{1}_S$ may have entries equal to 2 and must not be treated as Boolean; the signed count does not discard those terms.
5. **The height statistic is $h(\mathbf{1}_{\text{supp}(b)})$, not $h(b)$.** The nonlinear-height example on the three-element chain distinguishes the two.
6. **The weighted count uses the dual preorder.** Its source neighbourhoods are $D_\tau(i)$, not $U_\tau(i)$. This orientation is checked twice, in Proposition 5.1 and again in the support-restricted graph of Section 6.
7. **The zero-weight universal vertex remains present.** In the untrimmed Postnikov formula its factor is $f_{a_0}(0 + 1) = 1$, not $\binom{a_0-1}{a_0}$; the new 0-coordinate is included when applying the theorem, so the draconian total is n and not $n - 1$. Deleting that vertex would change both the total and the formula.
8. **Interior strictness is integral strictness.** After subtracting $\mathbf{1}$, the capacity $\mathbf{1} + \mathbf{1}_T$ produces $y(I) \leq |I \cap T| - 1$, not $y(I) \leq |I \cap T|$; likewise in Engine 2 the shift by $\mathbf{1}_R$ removes the private cube and leaves $y(J) \leq |N(J)| - 1$.
9. **The support-restricted graph need not be connected initially.** The isolated-vertex case has both counts zero; in the remaining case the universal right neighbour makes the graph connected, meeting the hypothesis of the imported duality. Correspondingly, in Engine 2 no empty simplex is ever passed to the nonempty-simplex formula: a resource with no nonprivate neighbour is handled before any polytope is formed.
10. **Degree vectors, not allocations.** The objects in bijection with maximal points are distinct full-allocation degree vectors. Counting maps or matchings with multiplicity would give the wrong theorem, which is also why the dynamic program of Section 14 must deduplicate.
11. **The multivariate step is not a weighted q -Ehrhart assertion.** It is the explicit two-value identity of Lemma 7.1, followed by ordinary univariate Ehrhart reciprocity along each fixed capacity ray.
12. **Capacities are independent variables before any negative specialization.** In Proof B reciprocity is established in all $n + 1$ variables first, and only then is v set to 0; there is no polytope of capacity -1 .
13. **Preorders, not just partial orders.** Privatization uses reflexivity and transitivity and never antisymmetry; genuine equivalence classes are allowed and retain their individual coordinates. Boundary cases with empty E , empty residual resource set and empty target sets are treated explicitly. The exhaustive verifiers include relations with genuine cycles.

15.3 What is established, and what is not

The arguments supplied here prove Theorems 1.1, 1.2 and 1.3 from the stated classical results, with no unproved additional combinatorial assertion and no hidden lemma. In that mathematical sense this is a complete proposed solution of the chosen conjecture, and its validity is not inferred from the finite computations.

The following separate questions are *not* answered by the computations or by this presentation: independent expert verification of every step; priority relative to unpublished or newly posted work; and a formal proof checked by a small trusted kernel. No such verification or priority certificate is claimed. This is an AI-assisted research draft, not independently refereed, not formalized in a proof assistant, and not represented as accepted or published mathematics. The literature checks were performed on 20 September 2026 and are documented in Appendix D

and `notes/source_audit.md`. The target is identified by both conjecture number and displayed formula, so a future change of source numbering need not make the statement ambiguous. It should be added that three separate drafts reaching the same two statements by three different routes is evidence of internal consistency, not of correctness; a shared misreading of the source's conventions would not have been detected by that agreement alone, which is why those conventions are restated explicitly in Section 1 and audited above.

15.4 Provenance of the source discrepancy

All three merged drafts recorded the same anomaly about the target's version metadata, each slightly differently, and the union is retained. The arXiv submission record displayed only v1, submitted 26 May 2026; the PDF's author date is 27 May 2026; and the retrieved HTML bears an internal date in August 2026 — one draft recording it as 24 August 2026. The three drafts therefore pin the target by arXiv identifier, section, equation number and displayed formula rather than by any of those dates, and the PDF's conjecture page was inspected visually as well as through parsed text. The relevant q -zeta statement agrees between all representations consulted.

Two further provenance points are kept verbatim from their original drafts. First, only the manifest, and not the full text of the earlier packages, was available when nonduplication was assessed. Second, the manifest's relevant entry is preserved word for word, with its SHA-256 digest and its source line numbers 960–969, in `notes/manifest_provenance.txt`; that file is the precise record of the nonduplication boundary, and it is the only place where the earlier entry is reproduced exactly.

16 Conclusion

The negative q -zeta evaluation is controlled by a small-coordinate part of the lattice-point poset, but its positive interpretation is global: maximal points of total n with a specified support. Proof A links the two views geometrically. In Engine 1 the support-restricted graph G_T does the work: its left draconian sequences encode maximal points allowed to use T , its right sequences encode interior points at capacities $\mathbf{1} + \mathbf{1}_T$, classical reciprocity converts the latter to negative polynomial values, and a two-entry interpolation table produces the same local weights as incidence inversion. In Engine 2 the work is done instead by privatizing the residual targets and by an Euler identity valid for every bipartite graph, which trades length for a smaller imported surface. Proof B links the two views algebraically, with no geometry beyond the one reciprocity statement: an auxiliary capacity is kept alive, then set to zero, and an involution on local power series moves the whole computation onto the maximal layer — which is also why it, and it alone, reaches every $[-r]_q$.

Together these yield the full Athanasiadis–Chapoton q -reciprocity formula, the stronger exact-support identity (1.9), its arbitrary-height version, and positive integral formulas at every negative q -integer. The three obstructions of Section 13 show that the cancellation is not an automatic property of pure coordinate downsets, of arbitrary assignment models, or of arbitrary capacities. It is the transitive ideal–filter structure of a preorder, with unit capacities, that makes the different descriptions agree.

A The finite Hall theorem used in Proof A

We recall the standard finite theorem [7], with its short inductive proof, in order to fix the exact matching convention. Let L, R be finite disjoint sets and let $N(A) \subseteq R$ be the neighbours of

$A \subseteq L$. There is an injective assignment $f : L \rightarrow R$ with $f(i) \in N(\{i\})$ if and only if

$$|A| \leq |N(A)| \quad (A \subseteq L).$$

Necessity is immediate. For sufficiency, induct on $|L|$; the empty case is trivial. If some nonempty proper $A \subset L$ has $|N(A)| = |A|$, match A into $N(A)$ by induction; for $B \subseteq L \setminus A$,

$$|N(B) \setminus N(A)| = |N(A \cup B)| - |N(A)| \geq |B|,$$

so induction matches $L \setminus A$ into $R \setminus N(A)$ and the two matchings combine. If no such nonempty proper A exists, choose $i \in L$ and a neighbour $r \in N(\{i\})$; after deleting i and r , every nonempty $B \subseteq L \setminus \{i\}$ still has at least $|B|$ neighbours, since originally it had at least $|B| + 1$. Apply induction and add the pair (i, r) . The singleton case is included directly by the existence of a neighbour.

Every use with integer multiplicities replaces a demand i of multiplicity a_i by a_i distinct copies. For any subset of copies, its cardinality is at most the total multiplicity of its set of types, while its neighbourhood is the neighbourhood of that set of types. This licenses every passage from vector inequalities to an ordinary finite matching in this document.

B A compact verification specification

A reimplementaion need not reproduce the Python organization. The essential checks are the following, all over the integers except the last.

B.1 Support, Boolean sums, and interior counts

For each enumerated preorder let $\mathcal{J}$ be its ideals and $\mathcal{F}$ its filters. For every $B, T \subseteq E$ compare

$$\begin{aligned} h(B) &= \#\{a \in \mathbb{N}^E : |a| = n, \text{supp}(a) = B, a(I) \leq |I| \ (I \in \mathcal{J})\}, \\ h(B) &\stackrel{?}{=} (-1)^{n-|B|} \sum_{S \subseteq E} (-1)^{|S|} \mathbf{1}[|B \cap I| + |S \cap I| \leq |I| \ (I \in \mathcal{J})], \\ \sum_{B \subseteq T} h(B) &\stackrel{?}{=} \#\{y \in \mathbb{N}^E : y(I) \leq |I \cap T| - 1 \ (\emptyset \neq I \in \mathcal{F})\}. \end{aligned}$$

The full-set bound makes the last enumeration finite, with $|y| \leq |T| - 1$; when $T = \emptyset$ and $n > 0$ its count is zero. The additional transport check compares the set of rank- n points above with

$$\left\{ \sum_{i \in E} e_{f(i)} : f : E \rightarrow E, i \leq_\tau f(i) \text{ for every } i \right\},$$

different assignments giving the same vector being counted only once.

B.2 The independent q -check

Enumerate all points a of total at most n satisfying the ideal bounds. At a fixed positive integer q , initialize $v_1(a) = q^{|a|}$ and iterate $v_{\ell+1}(b) = q^{|b|} \sum_{a \leq b} v_\ell(a)$; then $\sum_a v_{m-1}(a)$ is the positive value at $[m]_q$. Interpolate a polynomial of degree at most n through the $n+1$ nodes $m = 2, \dots, n+2$ using rational arithmetic and compare its value at $-1/q$ with $(-1)^n \sum_B h(B) q^{-|B|}$. At $q = 1$ use $[m]_q = m$. The interpolation must use exact rational arithmetic even at negative q -integers such as $[-2]_q = -3/4$ at $q = 2$.

B.3 Capacities, transfer, and negative evaluations

A relation is stored as n integer bit masks, bit j of row i being one exactly when $i \leq_\tau j$; transitivity is verified by requiring every bit of row j to be present in row i whenever bit j is present in row i . Ideals are generated by testing downward closure of every subset. This representation retains labels and does not collapse equivalence classes.

The capacity test implements (5.1) with the falling-factorial definition of the binomial coefficient, including negative upper arguments. The transfer test applies the triangular integral rule (9.4) to signed integer local polynomials. For a univariate polynomial the JSON coefficient array is in ascending powers, so $[0, 2, 9, 6, 1]$ means $2t + 9t^2 + 6t^3 + t^4$. Each example records the relation rows, the total lattice-point count, the number and full list of maximal points, the support polynomial and all four signed negative evaluations.

C Compact proof outlines for independent checking

Proof A compresses to the four-line chain displayed in Section 7, each line labelled with the ingredient it uses. Proof B compresses to the following five identities, starting from the count polynomial (5.1):

$$\begin{aligned}
\mathcal{E}_\sigma(-u, -v) &= (-1)^n \mathcal{E}_\sigma(u - \mathbf{1}, v - 1), \\
\sum_{a \in P_\tau} \prod_i \binom{a_i - u_i - 1}{a_i} &= (-1)^n \sum_{b \in \text{Max}(P_\tau)} \prod_i \binom{b_i + u_i - 2}{b_i}, \\
\sum_{a \in P_\tau} \prod_i [z^{a_i}] f_i(z) &= (-1)^n \sum_{b \in \text{Max}(P_\tau)} \prod_i [z^{b_i}] (\mathcal{T} f_i)(z), \\
q^a \begin{bmatrix} a - r - 2 \\ a \end{bmatrix}_q &= [z^a] \prod_{j=0}^r (1 - q^{-j} z), \\
\mathcal{T} \left(\prod_{j=0}^r (1 - t^j z) \right) &= \prod_{j=1}^r \left(1 + \frac{t^j z}{1 - z} \right).
\end{aligned}$$

The first line is interior-point translation plus classical reciprocity; the second is $v = 0$ with the dual preorder; the third is basis interpolation; the fourth is a finite Gaussian identity; the fifth is one rational substitution. For $r = 1$ the last expression has coefficient t in every positive degree, which is exactly the support statistic in the conjecture.

Engine 2 of Proof A compresses to: $\eta_\tau(B)$ is an alternating sum over the independent sets of a private-element bipartite graph (Lemmas 8.4 and 8.5); that alternating sum is $L_K(-1)$ for the polytope $K = [0, 1]^R + \sum_j \Delta_{A_j}^0$; Ehrhart reciprocity turns it into an interior count; a second application of Postnikov's theorem turns that into $\#(\sum_i \Delta_{N(i)} \cap \mathbb{Z}^C)$; and Hall's theorem identifies those lattice points with full-allocation degree vectors, which are in bijection with the maximal points of support B .

D Source and version ledger

The mathematical sources and the claimed additions are separated as follows. This ledger merges the three ledgers of the drafts named in Section 1.4; where they disagreed, the more cautious statement is kept.

Athanasiadis–Chapoton, arXiv:2605.26916v1.

Section 4.4, Conjecture 4.9 and equation (21), on printed page 14 of the retrieved PDF,

supply the target; Section 3.1 supplies the five-element example and its defining inequalities; Theorem 2.1 and Lemma 4.1 are the source’s own statement of Postnikov’s formula and of the draconian/filter identification. The source’s printed page 15 reports computational verification through size six and the value $-(2q+1)(q^2+4q+1)/q^5$. The version-date anomaly is recorded in Section 15.4. The PDF’s conjecture page was inspected visually as well as through parsed text.

Chapoton, arXiv:2402.11979v2, revised 29 January 2025.

Supplies the q -zeta concept, its conventions, and the general height-weighted definition; Appendix C develops the q -incidence algebra of which Section 3 is a written-out specialization. It is not claimed that interpreting q -zeta values through incidence powers is new. No general weighted reciprocity theorem is assumed from this paper: the negative specializations used here are derived independently, twice (Remark 9.5).

Postnikov, arXiv:math/0507163v1; IMRN 2009(6), 1026–1106.

Theorem 11.3, printed page 28 of the arXiv version, supplies the lattice-point formula, with its untrimmed full-simplex shift retained exactly; Definition 9.2, Lemma 11.7 and Corollary 11.8 supply the bipartite duality used only by Engine 1. The pages containing both were checked in the PDF page images. The passage of that paper explicitly called an “alternative semiproof” is not what is imported; the stated theorem and its main proof are the dependency. These are explicit imports, not part of any novelty claim.

Ehrhart–Macdonald reciprocity.

Used only for full-dimensional integral polytopes with all capacities positive integers, univariately along $s \mapsto sc$. Three references are retained: Macdonald [4] as the historical source, Beck–Robins [5, Chapter 4] as the textbook statement, and Beck–Develin [6] as a primary-source proof of the rational cone reciprocity from which the polytope case follows. The Macdonald publisher page was not accessible through the browsing tool used, and that bibliographic entry was cross-checked indirectly in research-paper references; no claim is made that the original Macdonald paper was read in full.

Hall, JAMS 10 (1935), 26–30.

The finite marriage theorem, also proved in Appendix A.

Dai–Hou–Liu–Thawinrak–Wang, arXiv:2608.16037v2, 27 August 2026.

Checked as related recent work. Its preorder application concerns duality for the support enumerator of *all* lattice points, which is a different enumerator from the maximal-support polynomial here; see the V versus V^* example of Section 12. It is not a proof dependency. A text search for “zeta” in the retrieved HTML returned no matches; that is a search observation, not a certificate of absence of related mathematics.

enumerative-combinatorics/preorder-polytope-reciprocity.

The earlier package [9] in this collection, cited in the three places listed in Section 1.5. It is credited as prior work, not imported as a lemma, and not audited here.

User-supplied manifest.tex, 20 September 2026.

Used only to delimit the target and to assess nonduplication, never as a proof source. Only the manifest, and not the full text of the earlier packages, was available for that comparison. The relevant entry is preserved verbatim with its digest and line numbers in notes/manifest_provenance.txt.

Status searches, 20 September 2026.

Targeted searches combined the source title or arXiv identifier with “Conjecture 4.9”, “ q -zeta”, “reciprocity” and “proof”; the source manuscript, Chapoton’s paper and the cross-polytope paper were inspected. The checked source still states the target as a conjecture, and no separate resolution was located. Some results were irrelevant or aggregated. This is evidence

about the sources checked, not a universal priority claim: an unpublished proof, an unindexed announcement, or the same result in other terminology could have been missed.

E Suggested manifest entry

The included file `manifest_entry.tex` contains a ready-to-adapt entry. Its scope is deliberately limited to the full q -refinement of Conjecture 4.9, the stronger support identity, the evaluations at every negative q -integer and the arbitrary-height variant. It does not repeat the earlier packages' other claimed conjecture resolutions, and its wording retains the research-draft status instead of representing this investigation as independently refereed. The three drafts merged here proposed three different directory paths, none of which matched the actual location of any of them; the entry now names one correct path.

References

- [1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026. <https://arxiv.org/abs/2605.26916v1>. Target: Section 4.4, Conjecture 4.9, equation (21), printed page 14; also Theorem 2.1, Lemma 4.1 and Section 3.1.
- [2] F. Chapoton, *On a q -analogue of the Zeta polynomial of posets*, arXiv:2402.11979v2, revised 29 January 2025. <https://arxiv.org/abs/2402.11979v2>. The q -incidence algebra is developed in Appendix C.
- [3] A. Postnikov, *Permutohedra, associahedra, and beyond*, International Mathematics Research Notices **2009**, no. 6, 1026–1106. DOI: 10.1093/imrn/rnn153. <https://arxiv.org/abs/math/0507163>. Theorem and lemma numbering in this draft follows the arXiv version: Definition 9.2, Theorem 11.3, Lemma 11.7 and Corollary 11.8.
- [4] I. G. Macdonald, *Polynomials associated with finite cell-complexes*, Journal of the London Mathematical Society (2) **4** (1971), 181–192.
- [5] M. Beck and S. Robins, *Computing the Continuous Discretely: Integer-Point Enumeration in Polyhedra*, second edition, Springer, 2015, Chapter 4. DOI: 10.1007/978-1-4939-2969-6.
- [6] M. Beck and M. Develin, *On Stanley's reciprocity theorem for rational cones*, arXiv:math/0409562v3, revised 4 August 2005, available at <https://arxiv.org/abs/math/0409562v3>. Only the lattice-polytope consequence is used here, with full dimensionality checked explicitly at each application.
- [7] P. Hall, *On representatives of subsets*, Journal of the London Mathematical Society **10** (1935), 26–30. DOI: 10.1112/jlms/s1-10.37.26.
- [8] Z. Dai, Q. Hou, Z. Liu, W. Thawinrak and H. Wang, *Counting Lattice Points in Minkowski Sums of Cross Polytopes*, arXiv:2608.16037v2, 27 August 2026. <https://arxiv.org/abs/2608.16037v2>.
- [9] *Reciprocity and Matrix Duality for Preorder Polytopes*, research manuscript, 19 September 2026, in this collection under `docs/reports/enumerative-combinatorics/` in the directory `preorder-polytope-reciprocity`. It proves Conjectures 7.2, 4.2 and 4.4 of [1] and the ordinary $q = 1$ case of Conjecture 4.9, together with multivariate reciprocity for independently weighted capacities. Cited as prior work in this collection; not imported as a lemma and not audited here.

- [10] *Manifest of mathematical research packages*, user-supplied `manifest.tex`, received 20 September 2026. Relevant entry: “Reciprocity and Matrix Duality for Preorder Polytopes,” source lines 960–969. The manifest is used only to delimit the present task, not as a proof source. A verbatim entry with its digest accompanies this archive in `notes/manifest_provenance.txt`.